网络转向：人文学科视角的变迁

The Network Turn: Changing Perspectives in the Humanities

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3 Culture Is Data

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3 文化即数据

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The network framework shapes how we interpret the world around us. Nothing is naturally a network; rather, networks are an abstraction into which we squeeze the world. Nevertheless, almost anything can be turned into a network, whether it be the interactions between characters in Shakespeare's plays, the dissemination of memes on Facebook, or the trade network implicated in the ancient Roman brick industry. To understand how such diverse topics can both be conceived of and analysed as networks, it is perhaps easiest to think about one of the most crucial types of network for the creation and dissemination of culture: communication networks. The simple act of communication functions as an entry point for the study of more complex processes such as the dissemination of information or the spread of cultural practices.

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网络框架塑造着我们理解周遭世界的方式。没有什么事物天然就是一个网络；相反，网络是一种抽象形式，我们将现实世界压缩到这种形式之中。尽管如此，几乎任何事物都可以转化为网络：莎士比亚戏剧中人物之间的互动、脸书上网络模因的传播，或古罗马制砖业所涉及的贸易网络，皆可如此。要理解这些如此不同的主题如何都能被构想为网络并加以分析，最容易的切入点或许是考察一种对文化的创造与传播至关重要的网络——交流网络。简单的交流行为，可以成为研究更复杂过程的入口，例如信息传播或文化实践的扩散。

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We are accustomed to thinking about communication as a network because of the infrastructures that mediate our interactions in the modern world: telephone networks, the World Wide Web, and online social networking platforms. We understand now more than ever that when communication happens through these mediums, it can be traced and measured. Investigations following Edward Snowden's leak in 2013 revealed that the US National Security Agency and its UK counterpart, GCHQ, had 'broadly compromised the guarantees that internet companies have given consumers to reassure them that their communications, online banking and medical records would be indecipherable to criminals or governments' (Ball, Borger, & Greenwald, 2013). Through such data these agencies were able to discover an individual's network of associations and communication patterns. But citizens are not subject only to network analysis in the service of national security: information provided by Christopher Wylie in 2018 confirmed that at least 87 million Facebook users were impacted by illegal data harvesting by Cambridge Analytica, who used this data about users' social networks to profile individual voters during the 2016 Brexit referendum in the UK and the presidential election in the USA in order to target them with personalised political advertisements (Laterza, 2018).

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我们习惯于将通信视为一个网络，因为现代世界中的各种基础设施——如电话网络、万维网和在线社交平台——都充当了我们互动的中介。如今我们比以往更清楚，当通信通过这些媒介进行时，是可以被追踪和测量的。2013年爱德华·斯诺登泄露信息后的调查表明，美国国家安全局及其英国对应机构GCHQ“严重破坏了互联网公司向用户提供的保障，这些保障旨在确保用户的通信、网上银行和医疗记录无法被犯罪分子或政府获取”（Ball, Borger, & Greenwald, 2013）。通过这些数据，这些机构能够发现个人的社交关系网络和通信模式。但公民并非仅受国家安全目的下的网络分析所影响：克里斯托弗·怀利在2018年提供的信息证实，至少有8700万Facebook用户受到了剑桥分析公司的非法数据收集行为的影响。该公司利用有关用户社交网络的数据，在2016年英国脱欧公投和美国总统选举期间对选民进行个性化政治广告投放（Laterza, 2018）。

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However, a digital trail is not necessary to reconstruct communication networks, and the data sets need not be big. Sociologists were reconstructing

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然而，无需依赖数字足迹来重建传播网络，且数据集也不需庞大。社会学家正是通过这种方式对传播网络进行了重建

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networks long before the data deluge. A classic example is Wayne Zachery's observations of a single karate club over three years, from 1970 to 1972, recording interactions between the thirty-four members that took place outside the club (Zachary, 1977). In this context, the members of the club constitute the nodes and the edges are the interactions that occurred between them outside the club: seventy-eight edges were recorded. Zachary used these observations to model the conflict (using the Ford–Fulkerson algorithm) that led ultimately to the club splitting. The emphasis on in-person interaction to model social structure has been appropriated more recently in literary studies, for example in the now much-cited (and much-critiqued) *Literary Lab* pamphlet 'Network Theory, Plot Analysis', in which Franco Moretti attempts to reconstruct the plot of *Hamlet* as a network graph wherein two characters share an edge when they exchange words. He uses the resulting network diagram to argue for the plot centrality of Horatio (Moretti, 2011).

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在数据泛滥之前很久，就已经存在网络。一个典型的例子是韦恩·扎卡里对一家空手道俱乐部三年的观察记录，时间跨度为1970年至1972年，他记录了34名成员在俱乐部外发生的互动情况（Zachary, 1977）。在此背景下，俱乐部成员构成节点，而节点之间的互动则表现为边：共记录了78条边。扎卡里利用这些观察结果来模拟导致俱乐部分裂的冲突过程（使用福特-富尔克森算法）。近期，以面对面互动来模拟社会结构的方法也被文学研究所采用，例如现在被广泛引用且备受批评的《文学实验室》手册《网络理论与情节分析》中，弗兰科·莫雷蒂试图将《哈姆雷特》的情节重构为一张网络图，其中两个角色在交流时共享一条边。他利用生成的网络图来论证霍拉提奥在情节中的中心性地位（Moretti, 2011）。

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Letters offer perhaps the most intuitive way of reconstructing analogue communication networks. Scholars in the field of epistolary studies have increasingly turned to network analysis and visualisation methods, and all of the authors of this book have worked on this material over the past decade. These efforts include projects on the underground communications between English Protestants during the Catholic reign of Mary and how they kept a community united whilst it was under systematic religious persecution (Ahnert & Ahnert, 2015); the way correspondence networks and scholarly institutions co-evolved into an incredibly efficient machinery for scientific coordination and discovery in early modern Europe (Weingart, under review); the infrastructure of the intelligence networks that underpinned Tudor domestic and foreign policy (Ahnert & Ahnert, 2019); and the role of mapping and visualisation technologies in understanding the intellectual communities of the Republic of Letters (Edelstein et al., 2017, Coleman, 2020). The aims of these various projects are manifold, but they are guided by the desire to find ways to make very large data sets more tractable and to reveal global patterns about how people communicated. The infrastructure of communication networks can be viewed in multiple dimensions, the two most obvious being geographical and social. On the social level, these projects are concerned with people who might bridge communities and geographies in ways that make the world smaller – diplomats, travelling scholars, merchants,

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这些研究提供了或许是最直观的方法，用于重建类似通信网络。书信研究领域的学者越来越依赖网络分析与可视化方法，本书的所有作者在过去十年中均致力于此类研究。这些研究包括：在玛丽统治时期英国新教徒之间的地下通信方式，以及他们如何在系统性的宗教迫害下保持群体团结（Ahnert & Ahnert, 2015）；早期现代欧洲中通信网络与学术机构如何共同形成高效的科学协作与探索机制（Weingart，待审）；支撑都铎王朝国内及外交政策的信息网络基础设施（Ahnert & Ahnert, 2019）；以及地图绘制与可视化技术在理解文人共和国知识社群中的作用（Edelstein等，2017；Coleman, 2020）。这些不同项目的目标多种多样，但其核心都是为了寻找方法，使庞大的数据集更易于处理，并揭示人们交流方式的全球规律。通信网络的基础设施可以从多个维度来看待，最明显的两个维度是地理与社会层面。在社会层面，这些研究关注那些能够连接不同社区与地区、让世界变得更小的人员——如外交官、游学学者、商人等。

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or adventurers – which can be discovered using certain combinations of computational measures described in Chapter 5. On the geographic level, scholars can use this data to understand the physical infrastructure that allowed these communications to happen, the veins and arteries of roads along which these communications travelled; what it can tell us about the efficacy of courier systems, early postal systems, and sea routes; or which people were in the same location at the same time.

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或冒险者——可以通过第5章中描述的某些计算手段组合来发现。在地理层面，学者们可以利用这些数据了解促成这些通信发生的物理基础设施，以及这些通信所经过的道路的脉络；这能告诉我们关于信使系统、早期邮政系统及海上航线的有效性如何；或者哪些人在同一时间处于同一个地点。

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More abstractly, such work can help us to think about how we can model the spread of information, news, ideas, and concepts across space and time. Yann Ryan's work on the development of the newspaper industry in the early modern period has shown how we can model the movement of individual news items across Europe, as in the case of a story from the *Moderate Intelligencer*, a London newsbook, datelined Hamburg, 21 July 1649, which narrates the war between the Polish and the Cossacks: 'As of old in the War of Alexander and the Romans, so lately in Ireland, and in this, it appears, that it's not the multitude that overcomes, but the wisdome and valour of men, of a very numerous Army, a few usually turns the Scale.' 'How could a writer of news', Ryan asks, 'based in Hamburg, writing about war in Poland, know about the wars in Ireland?' (Ryan, 2018). Building on research into the pan-European nature of the flow of news, Ryan's work seeks to show Britain and Ireland's place within that complex and organic system, how that system can be both local and international, and how it can work with efficiency despite the lack of central planning.

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更抽象地说，这类研究有助于我们思考如何模拟信息、新闻、理念与概念在空间与时间上的传播过程。Yann Ryan关于近代早期报业发展的研究表明，我们可以模拟单个新闻事件在欧洲各地的流动情况——例如《Moderate Intelligencer》这份伦敦新闻刊物在1649年7月21日寄往汉堡时报道的波兰与哥萨克之间的战争："正如亚历山大与罗马人之间的战争那样，最近在爱尔兰也如此；看来并非人数众多才能取得胜利，而是人的智慧与勇气，即使一支庞大的军队，也往往能扭转局势。"Ryan问道："一个居住在汉堡的新闻作家，如何能够了解波兰的战争情况？"（Ryan, 2018）。基于对新闻在全欧洲范围内流动特性的研究，Ryan的研究试图揭示英国和爱尔兰在该复杂而有机体系中的位置，以及该体系如何既具有地方性又具有国际性，又如何在缺乏中央规划的情况下仍能高效运作。

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Although parameters of space and time have taken on new dimensions in the era of digital communications, the question of how we model the dissemination of information remains constant. Facebook's research wing has examined the spread of memes on the social networking platform. In a 2016 article, Lada Adamic's team examined the dissemination and evolution of thousands of memes, collectively replicated hundreds of millions of times on Facebook, showing how the information undergoes an evolutionary process that exhibits several regularities, including a power-law (i.e. highly skewed) distribution of memes in variant frequencies. This research also uncovers how subpopulations on the social network 'can preferentially transmit a specific variant of a meme if the variant matches their beliefs or culture' (Adamic et al., 2016). This sense of preferential transmission

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尽管在数字通信时代，空间与时间的参数呈现出新的特征，但如何对信息传播进行建模的问题却始终存在。Facebook的研究部门对社交平台上的模因传播进行了研究。在2016年的一篇文章中，Lada Adamic的团队研究了数千个模因的传播与演变过程，这些模因在Facebook上被复制了数亿次。研究表明，信息经历了一个具有规律性的演变过程，包括模因以幂律分布方式在不同频率下出现。这项研究还揭示了社交网络中的亚群体如何根据特定模因与其信仰或文化的匹配程度，优先传播该模因的某种变体（Adamic等，2016）。这种优先传播现象表明……

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explains in part why micro-targeting proved such a powerful weapon in the 2016 presidential election and Brexit referendum, but also why research by Soroush Vosoughi, Deb Roy, and Sinan Aral has discovered that fake news disseminates faster and further than the truth. Their data set of rumour cascades on Twitter from 2006 to 2017 (around 126,000 rumours spread by approximately 3 million people) showed that the top 1 per cent of false news cascades diffused to between 1,000 and 100,000 people, whereas the truth rarely reached more than 1,000 people (Vosoughi, Roy, & Aral, 2018).

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部分解释了为何微目标策略在2016年总统选举和英国脱欧公投中成为如此强大的工具，同时也说明了为什么Soroush Vosoughi、Deb Roy和Sinan Aral的研究表明，假新闻的传播速度比真相更快、范围更广。他们关于2006至2017年间推特上谣言传播的数据集显示（约126,000条谣言由约3百万人传播），最顶尖的1%假新闻传播范围可达1,000至100,000人，而真相很少能触及超过1,000人（Vosoughi, Roy, & Aral, 2018）。

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It is not just information that is disseminated; people move too. In such a context people go from being nodes to being the connecting edges between places, as in the article 'A Network Framework of Cultural History' (and the accompanying video), which provides an overview of the movements of humanity over the past 2,600 years (Schich et al., 2014). The art historian Maximilian Schich and his collaborators on this project used Freebase to find 120,000 individuals who were deemed notable enough in their lifetimes that the dates and locations of their births and deaths were recorded, 'ranging from Solon, the Greek lawmaker and poet, who was born in 637 BCE in Athens, and died in 557 BCE in Cyprus, to Jett Travolta – son of the actor John Travolta – who was born in 1992 in Los Angeles, California, and died in 2009 in the Bahamas' (Abbot, 2014).[7] The team used that data to create a video that starts in 600 BCE and ends in 2012. Each person's birthplace appears on a map of the world as a blue dot and their place of death as a red dot. As a city becomes more important, more notable people die there, identifying these locations as cultural hubs. For example, the data suggests that Paris overtook Rome as a cultural hub in 1789. Schich has argued that the benefit of such a perspective is that 'historians tend to focus in highly specialised areas … but our data allow them to see unexpected correlations between obscure events never considered historically important, and shifts in migration' (Abbot, 2014).

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不仅仅是信息被传播；人们也在移动。在这种背景下，人们从节点转变为地点之间的连接边，正如《文化历史的网络框架》一文（及相关视频）所展示的：该文概述了过去2600年人类的活动情况（Schich等，2014）。艺术史学家Maximilian Schich及其项目合作者利用Freebase找到了120,000位在其一生中具有显著影响的个体，这些人的出生和死亡日期与地点都被记录下来。这些人物包括希腊立法者与诗人索伦，他于公元前637年在雅典出生，并于公元前557年在塞浦路斯去世；以及演员约翰·特拉沃尔塔之子杰特·特拉沃尔塔，他于1992年出生于加利福尼亚州洛杉矶，并于2009年在巴哈马群岛去世（Abbot，2014）。该团队利用这些数据制作了一部从公元前600年延续到2012年的视频。每个人的出生地以蓝色点显示在世界地图上，死亡地则以红色点显示。随着城市的重要性提升，更多知名人士在此去世，这些地点因此成为文化中心。例如，数据表明巴黎在1789年取代了罗马成为文化中心。Schich认为，这种视角的好处在于"历史学家往往专注于高度专业化的领域……但我们的数据使他们能够发现那些历史上从未被视为重要的obscure事件与移民趋势之间的意外关联"（Abbot，2014）。

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[7] Schich and colleagues address potential biases in their data in the supplementary material of their publication, including 'biographical, temporal, and spatial coverage; curated versus crowd-sourced data; increasing numbers of individuals who are still alive; place aggregation; location name changes and spelling variants; and effects of data set language'.

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Schich及其同事在论文的补充材料中指出了数据中可能存在的偏差，包括"人物背景、时间与空间范围；精选数据与众包数据的差异；仍存活的个体数量不断增加；地点汇总；地名变化及拼写变体；以及数据集语言的影响"。

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Such a framework is generalisable, offering different approaches for large-scale questions. If we conceptualise the individual people within this aggregate model of migration as carriers, we can imagine how researchers could use the framework Schich proffered to chart how patterns of travel and migration disseminate particular materials, technologies, practices, or concepts. The framework also shares important parallels with the work of the GLEAM project at Northeastern University, which has developed an epidemic forecaster that predicts how given diseases might spread by combining data on population, transport networks, the characteristics of disease, and possible responses (like travel restrictions and vaccination efforts). Using this information, the forecaster generates a simulation of disease propagation across the globe. The importance of this project lies in its ability to make informed suggestions about when and where to disseminate vaccines to minimise the impact of potentially devastating epidemics. During the COVID-19 pandemic in 2020 GLEAM provided early insights into the most effective mitigation strategies to slow the global spread of the disease.

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这种框架具有可推广性，能够为大规模问题提供不同的解决思路。如果我们将这种迁移总体模型中的个体人员视为载体，那么就可以想象研究人员如何利用该框架来描绘旅行与迁移模式如何传播特定物质、技术、实践或概念。该框架与东北大学GLEAM项目的工作具有重要相似性，后者开发了一种流行病预测工具，通过结合人口数据、交通网络、疾病特征以及可能的应对措施（如旅行限制和疫苗接种措施）来预测特定疾病如何传播。利用这些信息，该预测工具能够生成全球疾病传播的模拟结果。该项目的重要性在于它能够提供关于何时何地分发疫苗以最小化潜在致命疫情影响的建议。在2020年COVID-19疫情期间，GLEAM项目为寻找有效缓解策略以减缓疾病全球传播提供了早期见解。

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The predictive power of network-based models is a new horizon for cultural studies. In general we tend to study artefacts or practices already in existence. Of course we may soon find ways and uses for making predictions about the cultural arena of tomorrow. But prediction is equally important for thinking about existing culture where our data are scarce or missing, reframed as interpolation or computationally assisted historiographic guesswork. An example of this is Shawn Graham and Scott Weingart's research on the problem of 'equifinality' faced on archaeological sites: when a network is uncovered but the final shape of that network could have been reached by many potential means (Graham & Weingart, 2015). In the example studied, the nodes in the network were a series of bricks bearing the stamps of certain producers (carrying information concerning the estate on which the brick was made, the brick maker, the landowner, and the year) at separate sites, which pointed towards a trade network. Using agent-based modelling as a kind of counterfactual laboratory, the authors predicted the various possible processes that might have formed the networks apparent in the archaeological evidence. By comparing the networks that would have been predicted by Peter Fibiger Bang's model of the

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基于网络的模型的预测能力为文化研究开辟了新的领域。通常我们倾向于研究已经存在的 artifact 或实践。当然，我们或许很快就能找到方法，对未来的文化领域进行预测。但对于数据稀缺或缺失的情况，预测同样重要——我们可以将这些情况重新定义为插值或计算辅助的历史学推测。例如，Shawn Graham 和 Scott Weingart 关于考古遗址中"等价最终性"问题的研究：当某个网络被揭露时，该网络的最终形态可能通过多种潜在方式实现（Graham & Weingart, 2015）。在所研究的案例中，网络中的节点是一系列带有特定生产者印记的砖块（这些印记包含砖块生产地、制砖者、土地所有者及年份等信息），这些砖块来自不同的地点，表明存在贸易网络。作者利用基于代理的建模作为一种反事实实验室，预测了可能形成考古证据中可见网络的各种过程。通过对比 Peter Fibiger Bang 模型所预测的网络，……

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G02-P056-002 | 英文原文

Roman economy, which he calls the 'imperial Bazaar', against those observed in the archaeological materials, Graham and Weingart showed how scholars can employ counterfactual simulations to probe the limits of historical theories. In this case, the methods of prediction are not used to learn something new, but as a formalised thought experiment to better understand the implications of a theorised historical causal explanation. If the simulated results do not match the historical evidence, it is an indication that one's historical explanation needs further articulation.

中文译文 | G02-P056-002

格雷厄姆和温加特指出，他所称的"帝国市场"式罗马经济与考古材料中的现象不同。他们展示了学者如何通过反事实模拟来探索历史理论的边界。在这种情况下，预测方法并非用于获取新知识，而是作为一种形式化的思想实验，以更好地理解理论化历史因果解释的含义。如果模拟结果与历史证据不符，则表明该历史解释需要进一步明确。

G02-P056-003 | 英文原文（原段接续 / 分段）

The foregoing examples, despite the often complex nature or size of the data under consideration, might nevertheless be considered simple in that they constitute unipartite (or unimodal) networks: that is, networks with nodes of just one type. As we have argued elsewhere, however, the kind of networks humanists are interested in are often more complicated: 'We don't just have people connected to people or websites to websites, we've got people connected to institutions to authored works to ideas to whatever else, and we want to know how they all fit together' (Weingart, 2013). This is where we can employ what are known as multipartite (or multimodal) networks.

中文译文 | G02-P056-003

上述例子尽管涉及的数据往往规模庞大或性质复杂，但仍可认为相对简单，因为它们构成的是单部（或单模）网络，也就是只包含一种节点类型的网络。然而，正如我们在其他地方所论述的，人文学者关注的网络通常更为复杂："我们拥有的不只是人与人的连接，或网站与网站的连接，而是人与机构、机构与创作的作品、作品与观念以及其他种种事物的连接；我们想知道它们如何联系在一起"（Weingart，2013）。这时，就可以使用所谓的多部（或多模）网络。

G02-P056-004 | 英文原文（原段接续 / 分段）

The flavour network is a bipartite example. Sebastian Ahnert's research with collaborators at the Barabási Lab was inspired by the food-pairing hypothesis proffered by the chef Heston Blumenthal and the food scientist François Benzi: that ingredients will taste good together in a dish if they share chemical flavour compounds. Until their 2011 article, this hypothesis relied on anecdotal rather than large-scale quantitative evidence (Ahn et al., 2011). Thanks to the efforts of food scientists around the world, we now have extensive information available about the many chemical compounds responsible for giving different foods their distinctive smells and tastes. The two node types generated from that data for the bipartite network are the ingredients (such as garlic, chocolate, coffee, basil) and the individual chemical compounds. In order to show the relationships between a particular set of nodes, bipartite networks are usually transformed by a 'one-mode projection'. This means that the resulting network contains nodes of only one of either of the two sets. In this case, a one-mode projection of the ingredients was generated and edges were created between any foods that shared one or more compounds; the edges between those ingredients were

中文译文 | G02-P056-004

风味网络就是一个二部网络的例子。Sebastian Ahnert与Barabási实验室同事开展的研究，受到厨师Heston Blumenthal和食品科学家François Benzi所提出的食物搭配假说启发：如果若干食材具有共同的风味化学成分，将它们放在同一道菜中，味道就会很好。在他们2011年的论文发表之前，这个假说依据的是个别经验，而非大规模的定量证据（Ahn等，2011）。得益于世界各地食品科学家的努力，我们如今已经能够获得大量资料，了解赋予不同食物独特气味和味道的各种化合物。由这些数据生成的二部网络包含两种节点类型：食材（如大蒜、巧克力、咖啡、罗勒）和具体的化合物。为了展示某一组节点之间的关系，通常会通过"单模投影"转换二部网络。换言之，转换后的网络仅保留原先两组节点中的一组。在这个例子中，研究者生成食材的单模投影，并在任何两种共享一种或多种化合物的食材之间建立边；这些食材之间的边则

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G02-P057-002 | 英文原文

weighted by the number of compounds they shared. The resulting network was then used to test the food-pairing hypothesis by cross-referring it with a data set of 56,498 recipes. This analysis revealed that Western cuisines show a tendency to use ingredient pairs that share many flavour compounds, supporting Blumenthal and Benzi's hypothesis, but East Asian cuisines tend to avoid compound-sharing ingredients. This work shows how systems and hypotheses that do not seem at first sight to be about networks can be approached productively through that framework.

中文译文 | G02-P057-002

按照它们共享的化合物数量赋予权重。随后，研究者将所得网络与包含56,498份食谱的数据集交叉参照，用以检验食物搭配假说。分析发现，西方菜系倾向于搭配共享许多风味化合物的食材，这支持了Blumenthal与Benzi的假说；但东亚菜系却倾向于避免搭配具有共同化合物的食材。这项工作表明，一些乍看与网络无关的系统与假说，也可以通过网络框架得到富有成效的研究。

G02-P057-003 | 英文原文（原段接续 / 分段）

Where might this kind of approach be useful in the humanities? It is particularly pertinent where one has sources or data containing two different types of entities, and especially when one is a subset of (or somehow contained within) the other. For example, if we think of miscellanies – a collection of literary compositions or pieces by several authors assembled in a volume or book – we could create a bipartite network linking texts (node type one) with the miscellanies that contain them (node type two). So, in our toy network diagram (Figure 9), if text A appears in miscellanies X, Y, and Z, and text B appears in miscellanies Y and Z, then the bipartite network has connections A-X, A-Y, A-Z, B-Y, and B-Z. By then undertaking a one-mode projection of texts connected by miscellanies, we can see which texts appear together in such miscellanies (middle), or conversely, by projecting this as a network of miscellanies connected by shared texts, we can see how closely particular miscellanies are related to one another (bottom). Scaling up this idea, we can begin to see how an analysis of all catalogued Renaissance poetry miscellanies might enable us to understand the early modern poetic universe in different ways. It could tell us which poems most commonly appeared together, it might help us see which manuscripts were most similar in their contents, and it may uncover patterns of scribal circulation.

中文译文 | G02-P057-003

这种方法在人文学科的哪些领域可能有用？当资料或数据包含两类不同实体，尤其是一类是另一类的子集或以某种方式包含于另一类之中时，这种方法格外适用。例如，杂纂是将多位作者的文学作品或篇章汇编成一卷或一册的文集；我们可以构建二部网络，将文本（第一类节点）与收录这些文本的杂纂（第二类节点）联系起来。因此，在示例网络图（图9）中，如果文本A见于杂纂X、Y、Z，而文本B见于杂纂Y、Z，二部网络就具有A—X、A—Y、A—Z、B—Y和B—Z这些连接。随后，对通过杂纂相联的文本作单模投影，我们便能看到哪些文本共同出现在这类杂纂中（图中部）；反过来，将网络投影为通过共同文本相联的杂纂网络，就能看到特定杂纂之间关系的紧密程度（图下部）。将这个思路扩大应用，我们可以开始考察：对所有已经编目的文艺复兴时期诗歌杂纂作分析，如何有助于以不同方式理解近代早期的诗歌世界。它可以告诉我们哪些诗歌最常一同出现，帮助我们识别哪些写本的内容最相近，也可能揭示抄写传播的模式。

G02-P057-004 | 英文原文（原段接续 / 分段）

An example of this kind of bipartite thinking can be found in Richard Jean So and Hoyt Long's analysis of the publication networks of early twentieth-century poets. In their analysis, the two node types are poets and journals, and the fact of publication in the same journal is the edge linking one type of node to another. Their aim was the interrogation of the social dimensions of early twentieth-century poetry production in the United States, Japan, and China. 'The little magazines and independent coterie

中文译文 | G02-P057-004

这种二部思维的一个例子可以在理查德·让·索和霍伊特·朗对二十世纪初诗人出版网络的分析中找到。在他们的分析中，两种节点类型分别是诗人和期刊，而同一期刊的发表则作为连接一种节点与另一种节点的边。他们的目的是探讨美国、日本和中国的二十世纪初诗歌创作的社会层面。“小型杂志与独立团体”

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G02-P058-001 | 英文原文

bipartite network of texts and miscellanies:

中文译文 | G02-P058-001

文本与杂项的二部网络：

G02-P058-002 | 英文原文

texts **miscellanies**

中文译文 | G02-P058-002

杂项内容

G02-P058-003 | 英文原文

projected network of texts:

中文译文 | G02-P058-003

文本的投影网络：

G02-P058-005 | 英文原文

edge weight indicating
two shared miscellanies, Y and Z

中文译文 | G02-P058-005

表示两个共享杂项的边权重，Y和Z

G02-P058-006 | 英文原文

projected network of miscellanies:

中文译文 | G02-P058-006

杂项的投影网络：

G02-P058-007 | 英文原文

edge weight indicating
one shared text, A

中文译文 | G02-P058-007

表示一个共享文本A的边权重

G02-P058-008 | 英文原文

edge weight indicating
one shared text, A

中文译文 | G02-P058-008

表示一个共享文本A的边权重

G02-P058-009 | 英文原文

edge weight indicating
two shared texts, A and B

中文译文 | G02-P058-009

表示两个共享文本A和B的边权重

G02-P058-010 | 英文原文

Figure 9 A simple example of a bipartite network made up of nodes and miscellanies (top) projected as a network of texts (middle), and projected as a network of miscellanies (bottom). Diagram by the authors.

中文译文 | G02-P058-010

图9 一个二部网络的简单示例：由节点和杂项组成的二部网络（上图），以文本网络的形式呈现（中图），又以杂项网络的形式呈现（下图）。该图由作者绘制。

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G02-P059-002 | 英文原文

journals of the time were essential mechanisms for organizing creativity, collaboration, and the transnational diffusion of poetic styles' (So & Long, 2013: 157). Their work identifies key brokers in literary circles that include key writers (such as Amy Lowell and Countee Cullen in the network of American poets, 1924–5), and journals (such as *Shishin* in the network of Japanese journals, 1927–9).

中文译文 | G02-P059-002

当时的期刊是组织创作、合作以及诗歌风格跨国传播的重要机制（So & Long，2013：157）。这些期刊揭示了文学圈中的关键中介人，包括重要作家（如美国诗人网络中的艾米·洛威尔和库恩蒂·卡伦，1924–5年），以及期刊（如日本期刊网络中的《Shishin》，1927–9年）。

G02-P059-003 | 英文原文（原段接续 / 分段）

For humanities scholars unfamiliar with the network framework, the breadth and inclusivity of the foregoing examples may be surprising. Without the metaphorical and analytic power of networks, it is difficult to understand how the interactions of characters in *Hamlet* are in any way comparable to the dissemination of memes on Facebook, or the chemical compounds shared between food ingredients. What makes them comparable is an intellectual and methodological shift by which we abstract our objects of study into data points that can be entered into a database or spreadsheet. This does not imply some shared property intrinsic to each of the subjects under study, rather it implies the widespread utility of networks as a lens through which to view many aspects of our shared world.

中文译文 | G02-P059-003

对于不熟悉网络框架的人文学者来说，上述例子所展现的广泛性和包容性可能会令人惊讶。如果没有网络的隐喻性和分析功能，就难以理解《哈姆雷特》中人物之间的互动如何与 Facebook 上模因的传播或食品成分间的化学化合物相类似。它们具有可比性的是一种认知与方法上的转变——我们将研究对象抽象为可以输入数据库或电子表格的数据点。这并非意味着被研究对象具有某种共同的属性，而是表明网络作为一种视角，能够帮助我们理解我们共同世界中的诸多方面。

G02-P059-004 | 英文原文（原段接续 / 分段）

The idea of talking about cultural artefacts as data, metadata, or data sets is in itself a potentially controversial move as it seems reductive, ignoring complexity, ambiguity, and qualitative assessment. Moreover, as Johanna Drucker has argued, the etymology of the word 'data' is itself problematic in the context of the arts and humanities: it comes from the Latin *datum*, which means 'that which is given'. The word 'data' therefore carries with it the meaning of an observer-independent fact which cannot be challenged in itself. Drucker prefers instead to think in terms of 'capta': 'that which has been captured or gathered', a coinage designed to draw attention to the fact that the very act of capturing data is oriented by certain goals, done with specific instruments, and driven by a specific attention to a small part of what could have been captured given different goals and instruments (Drucker, 2011). In other words, it requires decisions and interpretations that introduce subjectivity into the process. Throwing out the word 'data' completely, however, seems unwise given its ubiquity and acceptance across disciplines, especially if one of our aims is to create discourses and practices that break down traditional disciplinary divides. It is perhaps best, therefore, to make sure that scholars from the arts and humanities are not

中文译文 | G02-P059-004

将文化艺术品视为数据、元数据或数据集的想法本身具有争议性，因为这种做法似乎过于简单化，忽视了其复杂性、模糊性以及定性评估。此外，正如乔安娜·德鲁克所指出，在艺术与人文领域，“数据”一词的词源本身也存在问题：它源自拉丁语“datum”，意为“被给予的事物”。因此，“数据”一词意味着一种独立于观察者的事实，这种事实本身无法被质疑。德鲁克更倾向于使用“capta”这一概念：“被捕获或收集到的事物”，这一术语旨在强调，数据的采集行为是由特定目标驱动的，通过特定工具进行，并且是出于对可能被捕获内容中一小部分的关注（如果目标与工具不同则情况会不同）（德鲁克，2011）。换句话说，这一过程需要决策和解释，从而引入主观性。然而，完全摒弃“数据”一词似乎并不明智，因为该词在各个学科中广泛存在并被接受，尤其是当我们的目标在于创造能够打破传统学科界限的论述与实践时。因此，或许最好确保艺术与人文领域的学者不会……

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G02-P060-002 | 英文原文

held captive to assumptions about what data is, and rather play an active role in applying a self-reflexive and theoretical lens to the act of data creation and capture and its subsequent processing, not just in the humanities and arts, but across the disciplines. Dan Rosenberg's work on the history of data zeros in on the distinctions between data, fact, and evidence. He writes, 'facts are ontological, evidence is epistemological, data is rhetorical,' which means that 'the existence of a datum has been independent of any consideration of corresponding ontological truth' (Rosenberg, 2013: 18). Rosenberg's formulation helps us to see that data are what we determine them to be, but we do need to take responsibility to define them clearly.

中文译文 | G02-P060-002

他们被对"数据是什么"的假设所束缚，而应在运用自省与理论视角来审视数据的生成、捕获及其后续处理过程中发挥积极作用，这不仅适用于人文学科与艺术领域，也适用于各学科领域。丹·罗森伯格关于数据历史的研究着重于数据、事实与证据之间的区别。他写道："事实属于本体论范畴，证据属于认识论范畴，而数据则属于修辞学范畴"，这意味着"数据的存在与相应的本体论真理无关"（罗森伯格，2013：18）。罗森伯格的论述帮助我们认识到，数据是我们所定义的，但我们必须明确界定它们。

G02-P060-003 | 英文原文（原段接续 / 分段）

Two key issues for network data are those of absence and 'bias' (broadly defined). The concern we hear repeated most often by our colleagues in the arts and humanities is that the incompleteness or complexity of their source material make it unsuitable for network analysis. The concern arises, in part, from a misconception about what 'scientific' data looks like. It is important to remember, however, that almost any network data set, even in the sciences, is incomplete in some sense. In whatever way we define the connection between entities and the entities themselves, we are limited by the scope of available data and by the assumptions made when defining the nature of the connections. While this of course will affect the results of any quantitative analysis, this incompleteness is less of a problem than it may seem at first, for three main reasons.

中文译文 | G02-P060-003

网络数据的两个关键问题在于缺失与"偏差"（广义定义）。我们在艺术与人文领域的工作伙伴们最常提出的担忧是，数据源的不完整或复杂性使其不适合进行网络分析。这种担忧部分源于对"科学"数据性质的误解。然而，重要的是要记住，几乎任何网络数据集，甚至是在科学领域，在某种程度上都是不完整的。无论我们如何定义实体与实体之间的关联，都受到可用数据范围以及定义关联性质时所做的假设的限制。虽然这当然会影响任何定量分析的结果，但这种不完整性实际上并不像最初看起来那么严重，原因主要有三个。

G02-P060-004 | 英文原文（原段接续 / 分段）

The first reason is that, particularly in the context of the humanities, the bias in the source data is often itself of interest to the scholar. When we look at cultural phenomena and artefacts, we look at them with particular well-theorised frames. However, for much data we do not have a well-thought-out, critical apparatus to describe the 'how', 'who', and 'what' of the data, so it is not clear whether we are analysing a particular phenomenon, some artefact of the collection and accumulation process, or our own world views. For example, in the case of archives of historic documents, network analysis can help the researcher to better understand the history of that archive by drawing attention to deliberate inclusions or omissions of material, perhaps motivated by a political or religious collection agenda, the confiscation of documents, or even the deliberate destruction or censorship of materials.

中文译文 | G02-P060-004

第一个原因是，尤其是在人文科学领域，源数据中的偏见本身往往也是学者感兴趣的焦点。当我们研究文化现象与文物时，我们会使用经过充分理论化的框架来观察它们。然而，对于许多数据而言，我们并没有完善的、具有批判性的分析工具来描述这些数据的"如何"、"谁"和"什么"，因此无法明确我们究竟是在分析某种现象、收集与积累过程中的某个文物，还是我们的自身世界观。例如，在历史文献档案的研究中，网络分析可以帮助研究者更好地理解该档案的历史，通过关注材料中被刻意包含或遗漏的部分来揭示可能由政治或宗教收藏目的、文件没收，甚至是有意破坏或审查材料所引发的种种情况。

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G02-P061-002 | 英文原文

Such events in the history of an archive are detectable precisely because of the effects they have of distorting the results of the quantitative analysis. We might be able to extrapolate similar kinds of lessons for survival or loss of archaeological evidence. And more generally, it could help in the development of a consistent scholarly apparatus for understanding the peculiarities of digital archives and databases. Via source criticism, we still only know in the broadest strokes the implications of their formation and shape for the research we undertake. Articulating humanities network data in these terms, as a perspective rather than a 'bias' – which implies some underlying ground truth – will be a grand challenge for the next generation of researchers in this area.

中文译文 | G02-P061-002

档案历史中的此类事件能够被识别，正是因为它们会扭曲定量分析的结果。我们或许可以推导出类似的经验，以理解考古证据的保存或丢失情况。更普遍地说，这有助于建立一种一致的学术体系，从而理解数字档案与数据库的特性。通过资料批判，我们仍然只能大致了解这些档案的形成及其对我们研究的影响。以这种方式来阐述人文学科网络数据，作为一种视角而非"偏见"——即暗示着某种潜在的真实基础——将是对未来该领域研究人员的一大挑战。

G02-P061-003 | 英文原文（原段接续 / 分段）

The second reason is that, even in an incomplete network, the relative importance of nodes and edges according to a given network measurement can still yield meaningful results. Every network data set analysed in network science, be it a mobile phone network, a network between genes, or an online social network, is incomplete or limited by selection. But the limited scope in which the data is collected does not prohibit us from examining the relative importance of nodes or edges in the context of that scope, as long as we are aware of the factors that may shape our results.

中文译文 | G02-P061-003

第二个原因是，即使网络不完整，根据特定网络测量数据所得到的节点与边的相对重要性仍然能够产生有意义的结果。网络科学中分析的每一个网络数据集——无论是手机网络、基因网络还是在线社交网络——都是不完整的或受到筛选限制的。但数据收集的范围有限并不妨碍我们在该范围内研究节点或边的相对重要性，只要我们知道可能影响结果的因素。

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The third and most important reason why the incompleteness of networks does not pose an insurmountable problem is that the results of the quantitative analysis do not serve as a final result, but as a starting point for further, more detailed inquiry in the vein of traditional scholarship in the arts and humanities. In Chapter 6, we provide a guide through what this might mean practically for the research process.

中文译文 | G02-P061-004

网络不完整的第三个也是最重要的原因在于，定量分析的结果并非最终结论，而是作为艺术与人文领域传统学术研究中进一步详细探究的起点。在第六章中，我们将阐述这在实际研究过程中可能意味着什么。

G02-P061-005 | 英文原文（原段接续 / 分段）

The message that networks do not need to be complete should not be used to discount the problem of inconsistent sampling or poor data gathering. But network analysis can be part of the answer if it is employed as a tool for characterising the contours of a given archive or corpus, thereby helping to show how the data with which we work so often reinscribes structural bias. This raises an important issue when understanding and constructing cultural objects as data. The big-data approach to culture offers many opportunities, as discussed earlier, but with it comes a duty of care that is at once technical and ethical (Weingart, 2014a). These two aspects are intimately intertwined.

中文译文 | G02-P061-005

不必认为文献总集必须完整，这一观点不应被用来忽视样本不一致或数据收集不足的问题。但如果将网络分析作为描述特定档案或文献总集特征的工具，它就可以成为解决之道的一部分，从而帮助揭示我们所使用的数据如何重新塑造结构性偏见。在理解与构建文化对象作为数据时，这便提出了一个重要问题。正如前文所述，大数据方法为文化研究提供了许多机会，但随之而来的是一种技术性与伦理性的责任要求（Weingart, 2014a）。这两个方面紧密交织在一起。

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G02-P062-002 | 英文原文

The technical duty of care is often called ‘data cleaning’, a term used to describe the process of detecting and correcting (or removing) corrupt or inaccurate elements in a given data set. Issues within a single data set or database might be created by misspellings during data entry, missing information, or other invalid data. The term, however, is not without problems, as has been discussed in the provocatively titled piece ‘Against Cleaning’ (Rawson & Muñoz, 2019). Rawson and Muñoz’s point is that the metaphor of cleaning is problematic because it elides the burden of labour and intellectual contribution that it actually represents. They contend that: ‘Data cleaning is a consequential step in the research process that we often make opaque by the way we talk about it. The phrase “data cleaning” is a stand-in for longer and more precise descriptions of what people are doing in the initial phases of data-intensive research.’ The lack of discussion around such practices, they argue, has increased mistrust of quantitative approaches in the arts and humanities. The only way to deal with this is to begin talking about the labour of cleaning and to communicate its significance as an intellectual contribution.

中文译文 | G02-P062-002

技术层面的责任关怀通常被称为“数据清洗”，这一术语用于描述检测并修正（或删除）特定数据集中存在的不准确或损坏元素的过程。单个数据集或数据库中的问题可能由数据输入时的拼写错误、信息缺失或其他无效数据引起。然而，这一术语并非没有问题，正如《反对清洗》一文中所讨论的那样（Rawson & Muñoz, 2019）。Rawson和Muñoz指出，使用“清洗”这一比喻存在问题，因为它忽略了实际上所涉及的劳动与智力贡献。他们认为：“数据清洗是研究过程中的一个重要步骤，但我们的表述方式往往使其变得模糊不清。‘数据清洗’这一说法只是对数据密集型研究初期阶段人们所做工作的更详细、更精确的替代说法。”他们认为，由于缺乏对这类实践的讨论，人们对艺术与人文学科中的定量研究方法产生了不信任。解决这一问题的方法就是开始谈论清洗工作所涉及的劳动，并明确其作为智力贡献的重要性。

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One example is the preparation Ruth and Sebastian Ahnert undertook for the Tudor Networks of Power project. Their primary data set was a digitised archive, State Papers Online (SPO), which combines scans of the British State Papers archive with the Calendar descriptions of their contents. It provides metadata for each document that, in the case of the letters on which they were working (numbering circa 132,000 items for the Tudor period), contains valuable relational information including: name of sender, name of recipient, date of composition, place of writing, unique document identifiers, and a content description. The author, recipient, and place fields in particular needed cleaning. In the case of the author and recipient fields, the names needed disambiguation and de-duplication, for a number of reasons: variant spellings of early modern names, letters addressed to a titular office rather than a named individual (e.g. the Archbishop of Canterbury), changing office holders, changing titles over a person’s lifetime as they accrued honours and offices, and women’s names changing due to marriage. The complexity of the sender and recipient metadata fields meant that although 37,101 unique name entities were initially extracted, there were in fact only 20,656 unique correspondents. Place names had to be dealt with in a similar

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一个例子是Ruth和Sebastian Ahnert为“Tudor Networks of Power”项目所做的整理工作。他们的主要数据来源是一个数字化档案——State Papers Online (SPO)，该档案包含了英国国家文件档案的扫描件以及相关内容的目录说明。它为每份文件提供元数据，对于他们正在研究的信件而言，约有132,000份文件，其中包含有价值的关系信息，包括：发件人姓名、收件人姓名、撰写日期、书写地点、独特的文件标识符以及内容描述。特别是发件人和收件人的字段需要清理。对于发件人和收件人的名称，由于多种原因需要进行去重和澄清：早期现代名称的拼写差异、信件寄给某个机构而非特定个人（例如坎特伯雷大主教），职位变化，一个人因获得荣誉和职位而变化的头衔，以及女性名字因婚姻变化。发件人和收件人元数据字段的复杂性意味着，虽然最初提取了37,101个独特的姓名实体，但实际上只有20,656个独特的通信对象。地名也需要以类似方式处理。

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G02-P063-002 | 英文原文

way and geo-coordinates added. This effort led to the development of two custom data-cleaning tools. In addition, the curation of those data fields involved eighteen months of work (Hyvönen et al., 2019; Ahnert & Ahnert, forthcoming: chapter 1). This is a considerable workload that needed to be undertaken before network analysis could even be contemplated and the results of quantitative analysis could be trusted. This work should not just be considered a prerequisite for network analysis or other quantitative approaches, however, but a scholarly object in its own right: a data set other researchers can put to many uses. A cultural shift in the humanities is still required before such work is given the scholarly credit it deserves.

中文译文 | G02-P063-002

添加了路径与地理坐标。这一工作促成了两种定制数据清洗工具的开发。此外，这些数据字段的整理工作耗时十八个月（Hyvönen等，2019；Ahnert & Ahnert，即将出版：第1章）。这是一项相当大的工作量，在进行网络分析或验证定量分析结果之前，必须完成这项工作。这项工作不应仅被视为网络分析或其他定量方法的先决条件，而应被视为一项独立的学术成果——一个可供其他研究人员广泛使用的数据集。在此类工作获得应有的学术认可之前，人文科学领域仍需发生一场文化变革。

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A vital by-product of this labour is that during the process researchers become intimately acquainted with the shape of their data, its strengths, weaknesses, and biases. This is where the ethical duty of care comes in. Lauren F. Klein's MLA paper on 'Distant Reading after Moretti' argued that we need to look at our data to understand why 'distant reading does not deal well with gender, or with sexuality, or with race.' She suggests that we need data sets that:

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这项工作的一个重要副产品是，在过程中研究人员能够深入了解数据的特征、其优势、劣势和偏差。这就是伦理责任发挥作用的地方。Lauren F. Klein在MLA论文《Moretti之后的“远距离阅读”》中认为，我们需要审视数据，以理解为何“远距离阅读在处理性别、性取向或种族问题时表现不佳”。她建议我们需要那些能够反映这些问题的数据集。

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perform the work of recovery or resistance. An example: the corpus created by the Colored Conventions Project, which seeks to recover and aggregate evidence that documents the Colored Conventions of the nineteenth-century United States; these were organizing meetings in which Black Americans, both fugitive and free, came together to strategize about how to achieve social and legal justice. By making this corpus available for others to download, the CCP opens up the project of distant reading to texts beyond quote 'representative' samples, which tend to reproduce the same inequities of representation that affect our cultural record as a whole. (Klein, 2018)

中文译文 | G02-P063-005

执行恢复或抵抗的工作。例如：由"有色契约项目"所创建的文献总集，该项目旨在收集并整理能够证明19世纪美国有色契约的证据；这些会议是黑人美国人——无论是逃亡者还是自由人——聚集在一起，共同商讨如何实现社会与法律正义的策略。通过向他人提供该文献总集的下载权限，CCP使"远程阅读"项目能够涵盖那些并非"代表性"样本的文本，而这些文本往往会重复那种影响我们整体文化记录的不平等呈现方式。（Klein, 2018）

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Attending to the biases created in the creation of the archives with which we work allows us to address them. Networks can be part of that process, helping us to describe the data we have, and therefore allowing us to see where gaps and skews are present and to think critically about the ways in which they may be addressed.

中文译文 | G02-P063-006

关注我们在整理档案过程中产生的偏差，有助于我们应对这些偏差。网络可以成为这一过程的一部分，帮助我们描述所拥有的数据，从而让我们看清存在哪些缺失和偏差，并批判性地思考如何解决这些问题。

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By thinking about the culture around us as data we therefore not only open up opportunities for new analytical processes, we also identify an important way in which arts and humanities scholars can contribute critically to the ways we theorise the construction of data sets. The ways that this community is accustomed to thinking about these objects of study offers a different perspective on the status of data, its capture, preparation, shape, strengths and weakness, as well as the duty of care required to ensure it is fit for purpose. As the next chapter argues, one of the ways we can interrogate our data is through the lens of visualisation.

中文译文 | G02-P064-002

将我们周围的文化视为数据，这不仅为新的分析方法提供了机会，也揭示了艺术与人文学者如何以关键方式对数据集的构建理论做出批判性贡献。该群体习惯于思考这些研究对象的思维方式，为我们理解数据的性质、其采集、处理、形态、优势与劣势，以及确保数据适用于特定用途所需的关怀措施提供了不同的视角。正如下一章所论述的，我们审视数据的一种方式就是通过可视化手段来实现。

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4 Visual Networks

Visual networks are compelling, but are they effective? The conventional network graph of node and edge (points connected by lines) makes it possible to convey a tremendous amount of information all at once, in one view. Networks express an internal logic of relationships between entities that is inherently intuitive. But they also lack an explicit external spatial referent, whether the latitude and longitude of cartography, the scale and sequence of a timeline, or the categories and measures that mark the x-y axis of a statistical graph. They can represent relationships at human visible scale, or scale to dizzying, visually indecipherable complexity. The unbounded rhizomatic structure of the network has a malleable, infinitely re-orderable form that communicates in a way that is distinct from almost all other diagrams of data. This chapter addresses how that malleability can be harnessed for research and communication.

中文译文 | G02-P065-002

第4章 网络的可视化

网络可视化引人注目，但是否有效？传统网络图使用节点与边，即以线连接的点，能够在同一个视图中同时传达大量信息。网络表达了实体之间关系的内部逻辑，这种逻辑具有直观性。不过，网络图也缺乏明确的外部空间参照：它不像地图具有经纬度，不像时间轴具有尺度和先后次序，也不像统计图以横、纵坐标标明类别与度量。网络既能在肉眼可辨的尺度上表现关系，也可能扩展到令人眼花缭乱、视觉上无法辨识的复杂程度。网络不受边界限制的根茎式结构，具有可塑且能够无限重新排列的形态；这种形态传达信息的方式，几乎有别于所有其他数据图示。本章讨论如何将这种可塑性用于研究与交流。

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The sketch presented in Figure 10 is one of many drafts leading up to Mark Lombardi's work, 'BCCI-ICIC & FAB, 1971–91 (4th Version)', with which we opened this book. It is a rough, incomplete, marked-up draft with nodes outlined in blue, connections crossed out, and empty spaces awaiting more information. A comparison between this draft and the final version gives us some insight into the evolution of Lombardi's thinking. Externalising information and organising ideas spatially is not just a mode of presentation, it is an integral part of Lombardi's research into these complex financial systems. The story he finally shares has been refined through the process of sketching, editing, and redrawing. But it is not merely a cleaned-up version of the draft. It demonstrates Lombardi's command of visual rhetoric to lay out his argument for public consumption; to convince and persuade with his choice of colour, the quality of the line, the careful annotation, the apparent precision, the internal structure, and the overall shape. The fragments of evidence Lombardi originally collected on three-by-five notecards and arranged in notes and sketches do not speak for themselves; they are selected, edited, and enriched, using visualisation at stages throughout the process as a cognitive aide, a modelling activity, an argument, and a work of art in turn (see Tversky & Suwa, 2009 and Tversky, 2014). It is an iterative act of knowledge production that if executed in prose rather than in drawing would

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图10中呈现的草图是通往马克·隆巴迪作品《BCCI-ICIC & FAB, 1971–91（第4版）》之前的众多草稿之一。这是一份粗糙且不完整的草稿，节点用蓝色标出，连接线被划掉，空白处则等待更多信息补充。将这份草稿与最终版本对比，有助于我们了解隆巴迪思维的发展过程。将信息外化并以空间方式组织思想，不仅是呈现方式，更是隆巴迪研究这些复杂金融体系的核心部分。他最终分享的故事是通过草图绘制、编辑和重画的过程逐步完善的。但这并非草稿的简单整理版。它展现了隆巴迪在视觉修辞方面的能力，通过颜色选择、线条质量、细致注释、明显的精确性、内部结构以及整体形态，来阐述他的论点，以说服他人。隆巴迪最初在三乘五的卡片上收集的证据片段，通过笔记和草图的形式被筛选、编辑并丰富起来，在整个过程中，可视化功能既作为认知辅助工具、建模活动、论证方式，也是一件艺术作品（参见Tversky & Suwa, 2009和Tversky, 2014）。这是一种迭代的知识生产过程，如果以文字而非绘画形式呈现，则会是另一番景象。

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Figure 10 Lombardi, Mark (1951–2000): Untitled from the series BCCI, ICIC & FAB, 1996. Pen and ink and electrophotographic print on paper. New York, Whitney Museum of American Art. © 2019. Digital image.

中文译文 | G02-P066-001

–图10 隆巴迪，马克（1951-2000）：无标题作品，出自《BCCI、ICIC与FAB系列》，1996年创作。以钢笔和墨水及电子摄影法在纸张上打印。纽约，惠特尼美国艺术博物馆。© 2019。数字图像。

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Whitney Museum of American Art / Licensed by Scala. figure 1

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惠特尼美国艺术博物馆 / 由Scala授权发布。图1

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be very familiar to humanities scholars. This familiarity is an important point of reference when comparing hand-drawn visual networks like Lombardi's to those drawn with computer graphics software. Ben Fry, for example, points to Lombardi's work for lessons in a humanist view of data that uses visualisation as a story-telling medium requiring significant editing (Fry, 2016). Our contention is that handled in the right way, network visualisation can be both an important means of knowledge production and a powerful rhetorical tool.

中文译文 | G02-P067-002

这对人文学者来说非常熟悉。这种熟悉性在将伦巴迪的手绘视觉网络与用计算机图形软件绘制的网络进行比较时，是一个重要的参考点。例如，本·弗莱指出，伦巴迪的作品为从人文视角看待数据提供了启示，这种视角将可视化视为一种叙事媒介，需要大量的编辑工作（Fry, 2016）。我们认为，如果以正确的方式处理，网络可视化既可以是知识生产的重要手段，也可以成为强大的修辞工具。

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The role of the visual in the production of knowledge is the thesis of Johanna Drucker's *Graphesis*. She opens the book with a comparison of two network visualisations: Athanasius Kircher's *Ars magna sciendi* (1669) and Barrett Lyon's 'Web Traffic Visualisation'. Both are networks depicted as nodes and edges (see Figure 11). Kircher returns us to the diagrammatic approaches to knowledge discussed in Chapter 2. It is a development of Ramon Llull's system published in his *Ars generalis ultima* or *Ars magna*: a method of combining religious and philosophical attributes selected from a number of lists, designed to engage Muslims in debate and win them to the Christian faith. Kircher's diagram forms a complete bipartite graph, where every node of the first set is connected to every node of the second. The Lyon visualisation, part of his Opte Project, is a computer-generated network graph intended to map the Internet. Nodes represent individual servers and hosts, and edges represent the fibre, copper, or other connections between them. Drucker's purpose in juxtaposing these two graphs is to draw our attention to their relationship to information. Kircher's work, she argues, '*produces the knowledge it draws*': the effect of combination is to generate insights. By comparison, Lyon's network visualisation of web traffic 'only displays information'. This distinction is an important one when thinking about how we employ visualisation in our work. Off-the-shelf graphics packages are employed in research as a means both of generating insights and displaying observed phenomena. However, the word 'display' is a potentially misleading term. As Drucker puts it, certain visualisations 'act as if they are just showing us *what is*, but in actuality, they

中文译文 | G02-P067-003

视觉在知识产生中的作用是 Johanna Drucker 在《Graphesis》一书中提出的论点。她以对两个网络可视化图的对比作为开篇：Athanasius Kircher 的《Ars magna sciendi》（1669年）以及 Barrett Lyon 的 "Web Traffic Visualisation"。这两份作品都将网络以节点和边的形式呈现（见图11）。Kircher 再次提及了第2章中讨论的图表化知识研究方法。这是 Ramon Llull 在《Ars generalis ultima》或《Ars magna》中提出的系统的进一步发展——该系统通过从多个列表中选择宗教与哲学属性，旨在促使穆斯林参与辩论并使其接受基督教信仰。Kircher 的图表构成了一个完整的二部图，其中第一组中的每个节点都与第二组的每个节点相连。Lyon 的可视化图属于他的 Opte 项目，是一种用于绘制互联网的计算机生成网络图。节点代表单个服务器和主机，边则代表它们之间的光纤、铜线或其他连接。Drucker 将这两张图表并置的目的是让我们关注它们与信息之间的关系。她认为，Kircher 的工作"产生了它所呈现的知识"：这种组合的效果就是产生洞察力。相比之下，Lyon 的网络可视化图仅用于展示信息。这种区别在思考我们如何在工作中使用可视化技术时非常重要。现成图形软件包被用于研究，既可用于产生洞察力，也可用于展示观察到的现象。然而，"显示"一词可能具有误导性。正如 Drucker 所说，某些可视化图"似乎只是展示事实，但实际上并非如此"

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are *arguments made in graphical form*' (Drucker, 2014). We need to recognise the rhetorical power of visualisation if we are to remain alert to the hidden agendas of the graphical forms we encounter.

中文译文 | G02-P067-004

"是以图形形式呈现的论据"（Drucker，2014）。如果我们想要保持对所遇到的图形形式背后隐藏意图的警觉，就必须认识到视觉化的修辞力量。

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Figure 11 On the left, a combinatorial diagram entitled *Typus universalis, omnibus de quacunque re proposita questionibus formandis, aptus*, in Athanasius Kircher, *Ars magna sciendi* (Janssonius a Waesberg, 1669). Digital image. Max Planck Institute for the History of Science Library. CC BY SA 3.0. On the right, a visualisation from the Opte Project of the various routes through a portion of the Internet in 2005. Digital image. Wikimedia Commons.

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通用类型，适用于所有主题。图11：左侧为名为“问题形成之艺术”的组合图，由阿塔纳修斯·基尔彻于1669年出版（詹索尼乌斯与瓦斯贝格出版社）。数字图像。马克斯·普朗克科学史研究所图书馆。CC BY SA 3.0。右侧为Opte项目对2005年互联网某部分网络的各条路径的可视化图。数字图像。维基共享资源。

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We can begin to understand how arguments are made in graphical form by looking at a set of diagrams based on the networks chapter of Jacques Bertin's *The Semiology of Graphics* (Figure 12). Bertin defines a general theory of graphics as separate from both figurative representation and mathematics. While Bertin employs some problematic assumptions about both the certainty of data and the 'monosemic' system of graphical rhetoric, his work is nevertheless a crucial study for our thinking about different insights, and consequently the arguments that can be made through different diagrammatic layouts of network data.

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通过查看雅克·贝尔坦所著的《图形符号学》一书的网络章节中的一系列图表（图12），我们可以开始理解以图形形式表达论点的方式。贝尔坦将图形的一般理论定义为与形象表现和数学相分离的一种概念。尽管贝尔坦对数据确定性以及“单义性”图形修辞体系提出了一些有问题的假设，但他的研究对于我们理解不同图表布局所能够表达的论点仍具有至关重要的意义。

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The network, according to Bertin, is represented by three components: the line, the point, and the area. He specifies a number of possible layouts making use of those components including rectilinear, circular, irregular arrangement, and regular arrangement (see Figure 12). Each combination of components and arrangements highlights different aspects of the data and contributes to a different argument about the underlying data. The tabular view provides the basic source–target pairs. The 'irregular arrangement' is the familiar, decentred layout most often associated with visual networks. It does not spatially prioritise any node and yet, with a network this small, we can quickly assess that A, B, C, and E are more connected than the other nodes. The 'regular arrangement' explicitly sets A as the root node in a tree-like hierarchy. The top-to-bottom order of the horizontal planes of alignment is clearly intentional and insists on a particular reading of the graph, even if the significance is not made explicit. And whereas the 'regular arrangement' gives prominence to A, the circular layout emphasises the node with the most connections. The lines radiating out from C give it a visually significant role in the network even though all the nodes are aligned on a periphery. The rectilinear layout suggests a sequence to the connections and makes clear, with this particular data set, that E connects more nodes along the chain than any other node (which, as we see in Chapter 5, would mean it has the highest betweenness centrality). The parallel alignment diagram and the matrix reveal directionality and absence more explicitly than the network graphs because they both draw corresponding

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根据Bertin的观点，该网络由三个组成部分构成：线、点与区域。他列举了多种利用这些组成部分的布局方式，包括直线式、圆形、不规则排列及规则排列（见图12）。每种组件与布局的组合都能突出数据的不同方面，从而为关于底层数据的不同观点提供支撑。表格形式展示了基本的源-目标对。“不规则排列”是一种常见的以中心为基准的布局，通常与视觉网络相关。这种布局并不对任何节点进行空间上的优先排序，但鉴于该网络规模较小，我们可以迅速判断A、B、C和E与其他节点的连接性更强。“规则排列”则明确将A设为树状层次结构中的根节点。水平排列平面的从上至下的顺序显然是有意为之，它强调了对图的特定解读方式，即使其重要性并未明确说明。而“规则排列”突出A的重要性，圆形布局则强调连接最多的节点。从C向外辐射的线条使该节点在网络中具有重要的视觉作用，尽管所有节点都位于边缘位置。直线式布局暗示了连接的顺序，并且通过这一特定数据集可以清楚地看出，E在链条上连接了更多节点，比其他任何节点都多（正如我们在第5章中所看到的，这意味着它具有最高的中介中心性）。平行排列图与矩阵比网络图更清晰地揭示了方向性和缺失情况。

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connections between two fixed planes. Since these two layouts accommodate all possible connections, they draw our attention to where there are

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Figure 12 This figure shows a number of different ways to present the same network data. On the left is source–target pairs in a two-column table. The regular and irregular arrangement, circular, and rectilinear are all network layouts in Bertin's system. He distinguishes those node-link network graphs from the diagrams: parallel alignment and matrix. Diagram by the authors.

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图12展示了呈现相同网络数据的多种不同方式。左侧是一个两列表格，包含源节点与目标节点的对。常规排列、不规则排列、圆形排列和直线排列都是Bertin系统中的网络布局形式。他将这种节点-链接网络图与另一种图表区分开来：平行排列和矩阵式图表。该图表由作者绘制。

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none. Even with this simple overview, it quickly becomes apparent how much the arrangement of nodes influences our reading of a network.

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无。即便只有如此简单的概述，也很快就能看出节点排列对我们理解网络结构的影响有多大。

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What is useful about Bertin's work is that it gives us a point of reference for reading the rhetoric encoded in design decisions. For example, if we return to Lombardi, we can see Lombardi's drawings align with Bertin's rules and extend them in unique ways. Lombardi's networks are hand drawn with a fine black line on white paper, sometimes accented with red, with descriptive labels and notes. He distinguishes six different types of connection between nodes: 1) influence or control in one direction, 2) mutual connection, 3) financial connection, 4) sale or transfer of assets, 5) a blocked or incomplete transfer, and 6) the sale or spin-off of a property. Moreover, Lombardi uses the rectilinear arrangement of nodes to add a temporal dimension to the visual network. In some cases, instead of a one-to-one relationship between node and entity, an organisation (usually a financial institution) is represented as a horizontal line with a beginning and an end. These are the primary entities, depicted as chronologies rather than singular nodes. The differences in the quality of the lines (solid, dashed, broken, twisted) are difficult to see at a distance, making the scale of his work an important part of the experience of it (the final work is 129.9 cm x 349.9 cm). Stepping back from the work, it is possible to take it all in as a complete system. The labels and notes are not legible at that distance, though the lines and vertices are clear. Even if the temporality is not explicitly visible, the horizontal lines add another layered dimension to the connectedness. A distant reading complements the close reading. Standing next to the work, the different types of

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Bertin的工作具有实用价值，它为我们解读设计决策中所包含的修辞表达提供了参考点。例如，如果我们回到Lombardi的作品，可以发现他的图纸符合Bertin的规则，并以独特方式加以扩展。Lombardi的网络是用细黑线在白纸上手绘而成，有时还会用红色标注，并附有描述性标签和注释。他区分了节点之间六种不同的连接类型：1) 单向影响或控制；2) 相互连接；3) 财务联系；4) 资产销售或转让；5) 阻塞或不完整的转让；6) 资产的出售或分拆。此外，Lombardi利用节点的直线排列为视觉网络增添了时间维度。在某些情况下，节点与实体之间并非一对一关系，而是将机构（通常是金融机构）表示为具有起止点的水平线。这些主要实体以时间序列的形式呈现，而非单个节点。线条的粗细差异（实线、虚线、断线、扭曲线）在远处难以辨识，因此其规模成为欣赏作品的重要方面（最终作品尺寸为129.9厘米×349.9厘米）。从远处观察，可以将整个作品视为一个完整的系统。在如此距离下，标签和注释难以辨认，但线条和顶点却清晰可见。即使时间性无法明确显现，水平线也为连接关系增添了另一层维度。远观与细读相辅相成。站在作品旁，不同类型的连接方式……

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connection and the handwritten labels and notes are easy to see. It encourages the reader to follow paths through the network, shifting their own position in relation to the artwork.

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连接关系以及手写的标签和笔记都清晰可见。这促使读者沿着网络中的路径前行，从而改变他们与艺术作品之间的位置。

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Researchers have made attempts to automate Lombardi's layout, to capture it as a reproducible algorithmic process. Indeed, there is a significant body of work in graph drawing studies called 'Lombardi drawings' (e.g. see Duncan et al., 2010). Such attempts, however, ignore the real value of his drawings and what is at stake in their creation. Lombardi was interested in evidence-based *visual* argument (the 'narrative structure') about specific individuals and organisations engaged in particular types of relationship

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研究人员试图将伦巴迪的布局过程自动化，将其转化为可复现的算法流程。事实上，在图绘制研究领域存在大量关于"伦巴迪图形"的研究（例如参见 Duncan 等，2010）。然而，这些尝试忽视了其图形的实际价值以及创作这些图形所涉及的深层意义。伦巴迪关注基于证据的视觉论证——即与特定个人和机构之间特定类型关系相关的"叙事结构"。

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over time. And his sketching process, leading up to the final presented work, captures the process of formulating that argument. Conversely, software programs for visualisation use predefined graphics libraries where much of the decision-making is pre-baked into the tool. They therefore take out of the user's hands the experience of the roughly drawn sketch as part of a generative and exploratory thought process. Instead, it jumps right into the visually precise presentation that we associate with the argument. It is not only that computer graphics look precise, computer graphics software is encoded with assumptions about graphical meaning rooted in Bertin's theory of graphics as monosemic and unambiguous.

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随着时间的推移。他从初步草图到最终作品的形成过程，也反映了该论点的构建过程。相反，用于可视化处理的软件程序使用预定义的图形库，其中许多决策都已在工具中预先完成。因此，用户无法亲自参与对粗略草图的绘制过程，而直接进入与论点相关的视觉精确呈现方式。并非只是计算机图形看起来精确，这些软件还包含了基于Bertin图形理论中关于图形意义的假设，即图形是单义且明确的。

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Just because visualisation tools present results to us in ways that can be understood as deterministic does not mean we need to use them that way. But we do need to be aware of the assumptions encoded in the tools we use so that we can bend them to our own needs. What follows is a comparison of two network visualisation packages often used in the humanities: Palladio and Gephi. Palladio uses interaction, filtering, and complementary visualisation modes to facilitate the exploration of multidimensional data sets. Gephi uses dynamic simulation to apply layout algorithms to generate graph network visualisations. The comparison is guided by what we expect to gain from the visualisation of data. Isabel Meirelles uses the words 'descriptive' and 'argumentative' to distinguish between visualisation for the exploration of data and visualisation for the communication of findings. She is careful to note that these are not either/or categories, but two poles of a continuum. Descriptive visualisation, Meirelles writes, is a reflection of direct mappings of the data – the who, what, where, and when – whereas argumentative visualisation tends to the more speculative (Meirelles, 2018). This approach to visualisation foregrounds our relationship to the data to be visualised. Lombardi's BCCI project and some of his collected data are the common point of reference in our comparison to help reveal the advantages and potential pitfalls of using software-generated visual networks in the research process.

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仅仅因为可视化工具以可理解的方式呈现结果，并不意味着我们就必须如此使用它们。但我们必须意识到所使用工具中蕴含的假设，以便根据自身需求调整这些工具。以下对比了人文领域常用的两种网络可视化软件：Palladio和Gephi。Palladio通过交互、过滤和互补式可视化模式来辅助对多维数据集的探索；Gephi则通过动态模拟和布局算法来生成图网络可视化效果。这种对比是基于我们对数据可视化的预期目标而进行的。Isabel Meirelles使用"描述性"和"论证性"这两个词来区分用于数据探索的可视化与用于成果展示的可视化。她强调，这些并非非此即彼的分类，而是连续体上的两个端点。Meirelles指出，描述性可视化反映了数据的直接映射——即谁、什么、在哪里、何时——而论证性可视化则更偏向推测性（Meirelles, 2018）。这种可视化方式突出了我们与待可视化数据之间的关系。Lombardi的BCCI项目及其收集的数据是我们对比中的参考点，有助于揭示在研究过程中使用软件生成的视觉网络的优势与潜在问题。

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Palladio is a toolset rooted in an historical-cultural orientation to network analysis that is more art than science. It is the result of experimentation with custom-made visualisations designed to answer specific historical questions

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Palladio 是一套以网络分析为核心理念的工具集，它更属于艺术而非科学。它是通过定制可视化工具进行实验的结果，这些可视化工具旨在解答特定的历史问题。

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about the shape of political, social, and intellectual communities in the early modern period (Edelstein et al., 2017; Coleman, 2020). The techniques deemed successful in those experiments were distilled and combined into one suite of tools. The result is a basic force-directed bipartite network graph embedded within an interactive and interchangeable set of data views that includes a geospatial view, a tabular view, and filtering on temporal and categorical dimensions. The interactive tools are designed to work together to help reveal many types of connectedness by facilitating multiple different points of view combined with faceting and filtering on different dimensions of the data. It is assumed that no one view is complete, but that a more complete picture is understood as much by seeing what is missing as by seeing the extant data (see Rendgen & Weidemann, 2012: 155; Uboldi et al., 2013).

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关于近代早期政治、社会与知识界群体的形态（Edelstein等，2017；Coleman，2020）。那些在实验中取得成功的技术被整合在一起，形成一套工具集。最终得到的是一个基础性的力导向二部网络图，该图嵌入于一个交互式且可互换的数据视图中，包括地理空间视图、表格视图，以及针对时间和类别维度的筛选功能。这些交互式工具旨在协同工作，通过结合多种不同的视角以及对数据不同维度的细分和筛选，帮助揭示各种类型的关联关系。假设没有一种视图是完整的，但理解更全面的图景，既需要看到缺失的部分，也需要看到现有数据（参见Rendgen & Weidemann，2012：155；Uboldi等，2013）。

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An exploration of Lombardi's data set with Palladio, for example, might begin with a map of the known locations. The text printed on the side of Lombardi's *Untitled* sketch (Figure 10) indicates that BCCI was the brain-child of a Pakistani banker based in Karachi. The bank was controlled from Abu Dhabi, was incorporated in Luxembourg, was operated from London, Geneva, and the Cayman Islands, and was involved in joint ventures with local banks in Iran, Oman, France, and Switzerland. Since BCCI was, according to Lombardi, the legitimate front for the shadowy ICIC, exam-ining the locations of individuals he believed were involved could reveal potential connections that might spur further research. Adding the time span filter in Palladio would make the data Lombardi has about when the entities formed and dissolved, cross-referenced with the timeline of specific dates when particular individuals met or financial transactions were recorded, available to filter, making it possible to quickly and easily zero in on which activities are taking place where and at what time. And since Lombardi seems to have more information about the relationships between people and organisations than about locations, he can look at the same filtered subset of data in a graph view to reveal connections between individuals and organisations at those moments in time.

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例如，利用Palladio对Lombardi的数据集进行探索，可以从已知地点的地图开始。Lombardi的《无题素描》（图10）侧面的文字表明，BCCI是一位居住在卡拉奇的巴基斯坦银行家所创立的企业。该银行由阿布扎比控制，在卢森堡注册，在伦敦、日内瓦和开曼群岛开展业务，并与伊朗、阿曼、法国和瑞士的本地银行进行合资经营。根据Lombardi的说法，BCCI是ICIC这个隐秘机构的合法载体，通过研究他认为与相关人物有联系的地点，可以发现潜在的关联，从而推动进一步研究。在Palladio中添加时间范围筛选功能后，就可以利用Lombardi所拥有的关于各实体成立和解散时间的数据，与特定人物相遇或财务交易记录的具体日期进行交叉比对，从而快速轻松地确定哪些活动在何时何地进行。由于Lombardi似乎拥有更多关于人与组织之间关系的信息而非地点信息，他可以通过图表视图查看同一经过筛选的子集数据，以揭示那些时刻下个人与组织之间的关联。

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This example demonstrates how Palladio can be used for sketching, in the sense of exploring, or thinking through data as a cognitive aid. It aids cognition by combining visualisations with the counting of values and categories that can easily be grouped and regrouped, arranged and

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这个示例展示了帕拉迪奥如何被用于草图绘制、探索或思考数据，作为一种认知辅助工具。它通过将可视化与数值和类别的计数相结合，帮助认知过程，这些元素可以轻松地被分组和重新分组、排列……

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rearranged. The faceting and filtering combines with the visual description to enable a subtle interrogation of heterogeneous, multidimensional data sets by concept and duration rather than purely numeric values. It was designed to be used with historical and archival data, with the assumption of incompleteness and uncertainty in the sources. Successful use of Palladio therefore requires the researcher to make inferences based on the visualised data, but it leaves space for the expert to fill in the gaps in the many cases where their knowledge of the subject extends beyond the data collected. It minimises automatic animation in favour of explicitly selected actions, to encourage reflection as views are reoriented and data is manipulated. The application is there not to run an analytical model but to facilitate modelling, to create space for assessment and imagination.

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重新调整了布局。面化与筛选功能与视觉描述相结合，使得人们能够依据概念和持续时间来对异构、多维数据集进行细致分析，而非仅依赖数值数据。该工具旨在与历史及档案数据结合使用，并假设数据源存在不完整性与不确定性。因此，成功运用Palladio需要研究者根据可视化数据进行推断，同时也为专家提供了空间，以填补那些其专业知识超出所收集数据的范围的情况。它减少了自动动画的运作，转而鼓励明确选择的操作，从而促使人们在视角重新调整和数据被处理时进行反思。该应用的目的并非运行分析模型，而是促进建模过程，为评估与想象提供空间。

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Gephi, by comparison, was developed in the context of the interdisciplinary *e-Diasporas* project to analyse online migrant communities (Diminescu, 2012). The underlying data, referred to as an *e-diaspora corpus*, is a list of diasporic websites selected by the project authors. The tool reflects the project's interest in the power of network science and the desire to visually expose network topology. Its creators describe it as the Photoshop of network graphs and it uses computer simulations to show layout algorithms acting on network data (Bastian, Heymann, & Jacomy, 2009). The force-directed layouts are a class of algorithms used frequently in Gephi and the visual form with which readers will be most familiar because of their increasing ubiquity through high-volume use of Gephi and similar tools. In a force-directed layout, each element of the network is modelled as though guided by unseen forces, much as in the physical world: gravity, electromagnetic repulsion, material elasticity. Nodes may be thought of as electrons on a two-dimensional plane, forcing each other apart as they approach one another, and the edges may be conceived of as springs physically anchored between two nodes, drawing them together even as they repulsively push away. The plane has no meaning; there are no axes defining categories against which to measure node position. But as we are already aware, we do come to these resulting images with a set of assumptions about meaning. Gephi also allows users to apply a whole range of quantitative measures (discussed further in Chapter 5) and then colour or resize visual elements according to those attributes.

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相比之下，Gephi是在跨学科e-Diasporas项目背景下开发的，用于分析在线移民社区（Diminescu, 2012）。该文献总集所包含的数据是由项目作者筛选出的海外移民网站列表。该工具体现了项目对网络科学力量的关注，以及希望直观展示网络拓扑结构的愿望。其开发者将其称为网络图的Photoshop，并通过计算机模拟来展示作用于网络数据的布局算法（Bastian, Heymann, & Jacomy, 2009）。力导向布局是一种在Gephi中经常使用的算法类型，其可视化形式也因Gephi及类似工具的高频使用而变得广为人知。在力导向布局中，网络中的每个元素都被模拟为受到看不见的力引导，就像物理世界中的情况一样：重力、电磁排斥力、材料弹性等。节点可以被视为二维平面上的电子，当它们靠近时会产生相互排斥的作用；边缘则可以被视为固定在两个节点之间的弹簧，即使它们相互排斥，也会将两者拉近。这个平面没有实际意义，也没有定义节点位置的轴。但正如我们所知，我们对这些可视化结果其实带有一定的意义假设。Gephi还允许用户应用各种定量度量标准（详见第5章），并根据这些属性对视觉元素进行颜色或大小调整。

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G02-P075-002 | 英文原文

An analysis of Lombardi's data set with Gephi would make it possible to automatically generate a calculated layout based on the connectedness of nodes in his network. Layout models embedded in a software application are extremely powerful. It makes it easy to quickly apply a given model to lots and lots of data. However, by making it so easy to apply multiple different layouts to the same data set, small data problems can be overlooked and amplified. In the scientific context, where network analysis is based on equations and resulting measures, computer-generated graphics like those produced with Gephi have come under scrutiny for unnecessarily introducing ambiguity into a reading of network topology. The argument is that spatial arrangements, colours, shapes, and labels complicate measurements that are more clearly presented numerically. But the source of confusion in network graphics resulting from powerful visualisation tools like Gephi may have more to do with the fact that visualisation is too often learned rather than taught. Too often users allow the software to determine the layout or become enamoured of the visualisation and animation, losing sight of the intent to accomplish something on the spectrum of description to argument.

中文译文 | G02-P075-002

对Lombardi的数据集使用Gephi进行分析，可以自动根据网络中的节点连接关系生成计算出的布局。嵌入在软件应用中的布局模型非常强大，能够轻松地将某种模型应用于大量数据。然而，由于能够轻易地对同一数据集应用多种不同的布局，一些小的数据问题可能被忽视或放大。在科学领域，网络分析基于方程和测量结果，像Gephi生成的计算机生成图形因不必要的模糊性而受到质疑。其理由是，空间排列、颜色、形状和标签会复杂化原本以数值形式更清晰的测量结果。但由如Gephi这样的强大可视化工具导致的网络图形混乱，可能更多源于可视化往往是通过学习而非教学来掌握的。很多时候，用户让软件自行决定布局，或者沉迷于可视化和动画效果，从而忽略了实现描述到论证之间目标的目的。

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Tommaso Venturini, Mathieu Jacomy, and Pablo Jensen have begun to address this challenge by explicitly defining the strengths and weaknesses of what they have called visual network analysis (VNA) and providing guidelines for how to make effective use of algorithmic layouts to expose the topological structure of large networks (Venturini, Jacomy, & Jensen, 2019). The diagrammatic network layouts Bertin offered – which provide visual clarity through either symmetry or minimising overlapping connections – break down when networks reach the size of several hundred or thousands of nodes. The argument underlying VNA is that understanding the conceptual and mathematical underpinnings of the layout algorithms and choosing appropriate settings can result in effective, legible graph spatialisation. Even though the specific position of each node on the plane is not relevant, the density (or lack of density) of nodes is. And though the structure of the visualised graph does not strictly correlate with the underlying mathematical calculations of the network, the structure that emerges through polarisation of nodes, even in a very large network, functions effectively as exploratory data analysis (Tukey, 1977). Selectively highlighting variables in the data reveal patterns. Nodes can be sized according to degree measures (see Chapter 5);

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Tommaso Venturini、Mathieu Jacomy和Pablo Jensen已经开始应对这一挑战，他们明确界定了所提出的视觉网络分析方法的优缺点，并提供了如何有效利用算法布局来揭示大型网络拓扑结构的指导原则（Venturini, Jacomy, & Jensen, 2019）。Bertin提出的图形化网络布局方式——通过对称性或最小化重叠连接来实现视觉清晰——当网络规模达到数百或数千个节点时便不再适用。视觉网络分析的核心观点是：理解布局算法的概念与数学基础，并选择适当的参数设置，就能实现有效的、易读的图式表示。虽然节点在平面上的具体位置并不重要，但节点的密度（或缺乏密度）却至关重要。尽管可视化图的结构与网络的底层数学计算严格无关，但即使是在非常大的网络中，通过节点极化形成的结构也能有效作为探索性数据分析的工具（Tukey, 1977）。选择性地突出数据中的变量可以揭示模式。节点的大小可以根据度度量来确定（详见第5章）。

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brightness can be applied to continuous numerical variables and hue for categorical variables. And just as with the experience of Lombardi's drawing, here closeness and distance are both revealing: zooming in and out in a Gephi visualisation makes global and local patterns visible. Even if the end goal is not a statistical reading of the network, the statistical readings available through Gephi provide intellectual prompts to examine your data in new ways.

中文译文 | G02-P076-002

"亮度"可用于连续数值变量，而"色相"则适用于分类变量。就像 Lombardi 的图表一样，在这里，"接近度"和"距离"都具有揭示性：在 Gephi 可视化中放大或缩小视图，可以使全局与局部模式清晰可见。即使最终目标并非对网络的统计分析，通过 Gephi 获得的统计结果也能提供新的思路，帮助以不同方式审视数据。

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Lombardi's work, Palladio, and Gephi represent three different kinds of visual engagement with data that require and deliver different levels of artistry, control, and quantitative engagement, but all of them encourage an important process of exploration. Network visualisations are often critiqued as either overly reductive (in terms of content and context) or absurdly illegible (in terms of visual complexity), yet they bring an immediacy to our perception of information. Graphics as a system for the creation and presentation of knowledge requires our attention. Graphics can, and often are, used throughout a research process from sketch drawings that help us build intuitions about connectivity, to descriptive data visualisation, to visual arguments in a final publication. Though data analysis is widely considered synonymous with particular forms of visualisation these days, charts and graphs were not widely accepted until they were formalised in the mid-twentieth century when computing and, specifically, computational information processing was on the rise. Statisticians Jacques Bertin and John Tukey, whose careers spanned the latter half of the twentieth century, were both pioneering figures who solidified the role of visualisation in providing insights that would otherwise be missed in mathematical data analysis. Bertin anticipated that the automation computers provide in calculation, combined with the ability to transform and reorder via computer display, would enable visualisation of much larger data sets. He seemed confident that computer graphics would be the apotheosis of his goal to produce 'the simplest and thus most communicable image'. However, one could argue that the ideal Bertin imagined has gotten away from us.

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Lombardi的研究成果《Palladio》与Gephi代表了三种不同的数据可视化方式，每种方式都需要并提供不同层次的艺术性、控制性与量化处理，但所有这些方式都促进了重要的探索过程。网络可视化常被批评为过于简化（就内容和背景而言）或难以理解（就视觉复杂性而言），然而它们却使我们对信息的感知更加直接。作为知识创造与呈现的体系，图形设计值得我们关注。在研究过程中，从帮助我们建立连通性直觉的草图，到描述性数据可视化，再到最终出版物中的视觉论证，图形设计被广泛使用。尽管如今数据分析通常被视为特定类型可视化的同义词，但图表和图形直到20世纪中期计算机技术，尤其是计算信息处理技术兴起之后才被广泛接受。统计学家Jacques Bertin和John Tukey在20世纪后半期从事研究，他们都是推动可视化在提供数学数据分析中难以发现的见解方面发挥作用的先驱人物。Bertin认为，计算机在计算方面的自动化能力，加上通过计算机显示进行转换和重新排序的能力，将使得更大规模的数据集也能被可视化。他坚信计算机图形将是实现“最简单且最易于传播的图像”这一目标的终极方式。然而，有人可能会说，Bertin所设想的理想状态已经远去。

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Cultural phenomena that Bertin could not have anticipated have radically changed our relationship to data graphics. Our day-to-day familiarity with mapping services is a case in point. We not only accept but rely upon their cartographic conventions to reduce the complexity of the landscape in order to serve a particular purpose: wayfinding. We navigate from point to

中文译文 | G02-P076-004

贝坦无法预料的文化现象彻底改变了我们与数据图形的关系。我们日常对地图服务的熟悉正是如此。我们不仅接受这些地图规范，还依赖它们来简化景观的复杂性，以服务于特定目的：导航。我们从一点到另一点进行移动。

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G02-P077-002 | 英文原文

point, implicitly accepting the abstraction of the map because we have our embodied experience of walking, riding, and driving streets to fill in the gaps. We know that travel between two nodes is rich and layered; it is never a straight line. We have learned to reframe space in scale from a global view to the city streets with longitude and latitude and familiar geography as a guide. We trust these constantly shifting maps perhaps because we are so aware of their limitations or because we rely on their convenience. As a result, we allow ourselves to be led by them, unaware of or tacitly accepting their encoded intention. The intervention of software as an unseen hand in the construction of visualisations presents new challenges to our reading of them. When the direct connection between the author and the output is not evident in the form, we can be seduced into experiencing visualisation as objective. This is even more likely when the map layer is removed and what is left is a visualised network graph without a physical referent.

中文译文 | G02-P077-002

我们默认接受地图的抽象性，因为我们有在行走、骑行和驾驶街道时的具体体验来填补其中的空白。我们知道两点之间的旅行是丰富且复杂的，从来不是一条直线。我们学会了从全局视角到城市街道的尺度空间，并以经度和纬度以及熟悉的地理信息作为参考来重新构建空间。我们信任这些不断变化的地图，或许是因为我们清楚它们的局限性，或者因为依赖它们的便利性。因此，我们允许自己被它们引导，而不意识到或默认接受其所蕴含的意图。软件作为看不见的手在可视化构建中的介入，给我们的解读带来了新的挑战。当作者与输出结果之间的直接联系在形式上看不清晰时，我们可能会被误导，认为可视化是客观的。当地图层被移除，剩下的是没有物理参照物的可视化网络图时，这种情况更为常见。

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Spatialisation of nodes and links, vectors and vertices, without reference to scale or familiar geospatial markers is unsettling and confrontational. And yet, eschewing the Cartesian hand-holds of the map reveals patterns and movement that would otherwise be obscured or overshadowed by the expectations we bring to the geospatial framing of data. It focuses our attention differently – from topography to the line and its inherent relationships. The points, lines, and areas by themselves can be disorienting and challenge our assumptions about what is significant in the data. At the same time, our relationship to data has changed radically in this century. Not only is there much more of it, it is much more personal. As a result, we are better able to see the human-scale biases and assumptions behind the abstract and inherently fragmented visualisation of networks. Just as networks are always metaphorical, visual networks are as much about schema as they are a set of relationships and measurements. These opportunities and challenges of the network have been seized upon by artists, scholars, and, more recently, the developers of visualisation tools and libraries. The need to concretise the abstract has given way to the desire to communicate expressively now that we have extremely powerful tools with which to fulfil that desire. The challenge is to redefine the role of graphics in network analysis for the future of network science, design, and humanistic inquiry.

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节点与连接、向量与顶点的空间化呈现，无需参考规模或熟悉的地理空间标记，这种呈现方式令人不安且充满对抗性。然而，摒弃地图的笛卡尔式框架，能够揭示那些被我们针对数据地理空间框架所持有的预期所掩盖或遮蔽的模式与动态。它以不同的方式聚焦我们的注意力——从地形到线条及其内在关系。点、线和区域本身可能令人困惑，并挑战我们对数据中重要内容的假设。与此同时，本世纪我们与数据的关系发生了根本性的变化：不仅数据量大幅增加，而且更具个人性。因此，我们更能看到网络抽象且碎片化的可视化背后的人类尺度偏见与假设。正如网络总是具有隐喻性一样，视觉网络同样既涉及模式，也包含一系列关系与度量。艺术家、学者以及最近的可视化工具与库开发者都利用了这些机会与挑战。由于我们有了强大的工具来实现这一愿望，对抽象事物的具体化需求已让位于表达性的沟通需求。挑战在于重新定义图形在网络分析中的角色，以服务于网络科学、设计与人本主义研究的未来。

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Dan Rosenberg, for example, posits a different analytic goal of

中文译文 | G02-P077-004

例如，Dan Rosenberg提出了不同的分析目标

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visualisation: ‘to complicate rather than simplify, to open multiple avenues of inquiry, and, most importantly, to challenge the stability of underlying data, in fact or in principle’ (Rosenberg, 2016).

中文译文 | G02-P078-002

可视化：“旨在使问题更加复杂而非简单化，开辟多种研究途径，最重要的是，挑战基础数据的稳定性，无论是实际还是理论层面”（罗森伯格，2016）。

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The fact that dense and complex networks resemble neural connections in the brain, rhizomatic weed roots pulled from the ground, twenty-four hours of airline flight, or the exchange of information over the Internet, communicates something about the very real complexity we experience in our everyday lives at many different scales, which we need to interrogate. We propose that visualisation is not better than narrative argument or mathematical equations for communicating ideas, but that it provides an additional means of producing, exploring, and analysing information that has proven value in both the liberal arts and the sciences. When working with data, visualisation can help reveal gaps that narrative-based research practices do not expose, or uncover qualities in data sets that can easily be overlooked when we focus only on numbers and calculations. Moreover, it is rhetorically powerful. Visualisations of networks are unsettling because the graphical language used to produce them is not, in fact, precise. For these reasons, we argue that visualisation has a challenging and provocative role to play in the development of network approaches to culture, a role that needs to be fully integrated into an iterative and self-reflexive research process. In Part III, we outline the manoeuvres that make up that process.

中文译文 | G02-P078-003

那些密集且复杂的网络与大脑中的神经连接相似，从地面拔出的根状杂草根，24小时空中飞行，或是互联网上的信息交换，都揭示了我们在日常生活中所经历的真实复杂性——这种复杂性需要在不同层面上进行探讨。我们认为，可视化并不比叙述性论证或数学方程更优越于传递思想的方式，但它提供了一种额外的手段来生成、探索和分析信息，这在文学与科学领域都具有价值。在处理数据时，可视化有助于揭示基于叙述的研究方法无法发现的空白，或者发现那些仅关注数字和计算时容易忽视的数据集特征。此外，它在修辞上也非常有力。网络的可视化令人不安，因为用于生成这些图表的图形语言实际上并不精确。因此，我们认为可视化在发展网络式文化研究方法中具有挑战性和启发性的作用，这种作用需要完全融入迭代且自我反思的研究过程之中。在第三部分中，我们概述了构成该过程的各个步骤。

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Part III Manoeuvres

中文译文 | G02-P079-001

第三部分 操作策略

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Network analysis provides a set of manoeuvres: intellectual manoeuvres that refigure cultural objects in our minds as abstract systems of nodes and edges; mechanical manoeuvres that structure data and that navigate input versus output; and manoeuvres between a landscape of abstraction and research questions that are steeped in contextual information. These manoeuvres dismantle the binaries between distant and close reading, quantitative and qualitative analysis, the nomothetic and idiographic, and, in so doing, create new norms of practice and inquiry.

中文译文 | G02-P079-002

网络分析提供了一系列操作方式：一种是智力层面的操作，它将我们头脑中的文化对象重新构建为抽象的节点与边构成的系统；另一种是机械层面的操作，用于整理数据并判断输入与输出之间的关系；还有介于抽象景观与充满语境信息的研究问题之间的操作方式。这些操作方式打破了远距离阅读与近距离阅读、定量分析与定性分析、普遍规律与特殊案例之间的二元对立，从而创造了新的实践与探究规范。

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3 Culture Is Data

The network framework shapes how we interpret the world around us. Nothing is naturally a network; rather, networks are an abstraction into which we squeeze the world. Nevertheless, almost anything can be turned into a network, whether it be the interactions between characters in Shakespeare's plays, the dissemination of memes on Facebook, or the trade network implicated in the ancient Roman brick industry. To understand how such diverse topics can both be conceived of and analysed as networks, it is perhaps easiest to think about one of the most crucial types of network for the creation and dissemination of culture: communication networks. The simple act of communication functions as an entry point for the study of more complex processes such as the dissemination of information or the spread of cultural practices.

We are accustomed to thinking about communication as a network because of the infrastructures that mediate our interactions in the modern world: telephone networks, the World Wide Web, and online social networking platforms. We understand now more than ever that when communication happens through these mediums, it can be traced and measured. Investigations following Edward Snowden's leak in 2013 revealed that the US National Security Agency and its UK counterpart, GCHQ, had 'broadly compromised the guarantees that internet companies have given consumers to reassure them that their communications, online banking and medical records would be indecipherable to criminals or governments' (Ball, Borger, & Greenwald, 2013). Through such data these agencies were able to discover an individual's network of associations and communication patterns. But citizens are not subject only to network analysis in the service of national security: information provided by Christopher Wylie in 2018 confirmed that at least 87 million Facebook users were impacted by illegal data harvesting by Cambridge Analytica, who used this data about users' social networks to profile individual voters during the 2016 Brexit referendum in the UK and the presidential election in the USA in order to target them with personalised political advertisements (Laterza, 2018).

However, a digital trail is not necessary to reconstruct communication networks, and the data sets need not be big. Sociologists were reconstructing

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networks long before the data deluge. A classic example is Wayne Zachery's observations of a single karate club over three years, from 1970 to 1972, recording interactions between the thirty-four members that took place outside the club (Zachary, 1977). In this context, the members of the club constitute the nodes and the edges are the interactions that occurred between them outside the club: seventy-eight edges were recorded. Zachary used these observations to model the conflict (using the Ford–Fulkerson algorithm) that led ultimately to the club splitting. The emphasis on in-person interaction to model social structure has been appropriated more recently in literary studies, for example in the now much-cited (and much-critiqued) *Literary Lab* pamphlet 'Network Theory, Plot Analysis', in which Franco Moretti attempts to reconstruct the plot of *Hamlet* as a network graph wherein two characters share an edge when they exchange words. He uses the resulting network diagram to argue for the plot centrality of Horatio (Moretti, 2011).

Letters offer perhaps the most intuitive way of reconstructing analogue communication networks. Scholars in the field of epistolary studies have increasingly turned to network analysis and visualisation methods, and all of the authors of this book have worked on this material over the past decade. These efforts include projects on the underground communications between English Protestants during the Catholic reign of Mary and how they kept a community united whilst it was under systematic religious persecution (Ahnert & Ahnert, 2015); the way correspondence networks and scholarly institutions co-evolved into an incredibly efficient machinery for scientific coordination and discovery in early modern Europe (Weingart, under review); the infrastructure of the intelligence networks that underpinned Tudor domestic and foreign policy (Ahnert & Ahnert, 2019); and the role of mapping and visualisation technologies in understanding the intellectual communities of the Republic of Letters (Edelstein et al., 2017, Coleman, 2020). The aims of these various projects are manifold, but they are guided by the desire to find ways to make very large data sets more tractable and to reveal global patterns about how people communicated. The infrastructure of communication networks can be viewed in multiple dimensions, the two most obvious being geographical and social. On the social level, these projects are concerned with people who might bridge communities and geographies in ways that make the world smaller – diplomats, travelling scholars, merchants,

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or adventurers – which can be discovered using certain combinations of computational measures described in Chapter 5. On the geographic level, scholars can use this data to understand the physical infrastructure that allowed these communications to happen, the veins and arteries of roads along which these communications travelled; what it can tell us about the efficacy of courier systems, early postal systems, and sea routes; or which people were in the same location at the same time.

More abstractly, such work can help us to think about how we can model the spread of information, news, ideas, and concepts across space and time. Yann Ryan's work on the development of the newspaper industry in the early modern period has shown how we can model the movement of individual news items across Europe, as in the case of a story from the *Moderate Intelligencer*, a London newsbook, datelined Hamburg, 21 July 1649, which narrates the war between the Polish and the Cossacks: 'As of old in the War of Alexander and the Romans, so lately in Ireland, and in this, it appears, that it's not the multitude that overcomes, but the wisdome and valour of men, of a very numerous Army, a few usually turns the Scale.' 'How could a writer of news', Ryan asks, 'based in Hamburg, writing about war in Poland, know about the wars in Ireland?' (Ryan, 2018). Building on research into the pan-European nature of the flow of news, Ryan's work seeks to show Britain and Ireland's place within that complex and organic system, how that system can be both local and international, and how it can work with efficiency despite the lack of central planning.

Although parameters of space and time have taken on new dimensions in the era of digital communications, the question of how we model the dissemination of information remains constant. Facebook's research wing has examined the spread of memes on the social networking platform. In a 2016 article, Lada Adamic's team examined the dissemination and evolution of thousands of memes, collectively replicated hundreds of millions of times on Facebook, showing how the information undergoes an evolutionary process that exhibits several regularities, including a power-law (i.e. highly skewed) distribution of memes in variant frequencies. This research also uncovers how subpopulations on the social network 'can preferentially transmit a specific variant of a meme if the variant matches their beliefs or culture' (Adamic et al., 2016). This sense of preferential transmission

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explains in part why micro-targeting proved such a powerful weapon in the 2016 presidential election and Brexit referendum, but also why research by Soroush Vosoughi, Deb Roy, and Sinan Aral has discovered that fake news disseminates faster and further than the truth. Their data set of rumour cascades on Twitter from 2006 to 2017 (around 126,000 rumours spread by approximately 3 million people) showed that the top 1 per cent of false news cascades diffused to between 1,000 and 100,000 people, whereas the truth rarely reached more than 1,000 people (Vosoughi, Roy, & Aral, 2018).

It is not just information that is disseminated; people move too. In such a context people go from being nodes to being the connecting edges between places, as in the article 'A Network Framework of Cultural History' (and the accompanying video), which provides an overview of the movements of humanity over the past 2,600 years (Schich et al., 2014). The art historian Maximilian Schich and his collaborators on this project used Freebase to find 120,000 individuals who were deemed notable enough in their lifetimes that the dates and locations of their births and deaths were recorded, 'ranging from Solon, the Greek lawmaker and poet, who was born in 637 BCE in Athens, and died in 557 BCE in Cyprus, to Jett Travolta – son of the actor John Travolta – who was born in 1992 in Los Angeles, California, and died in 2009 in the Bahamas' (Abbot, 2014).[7] The team used that data to create a video that starts in 600 BCE and ends in 2012. Each person's birthplace appears on a map of the world as a blue dot and their place of death as a red dot. As a city becomes more important, more notable people die there, identifying these locations as cultural hubs. For example, the data suggests that Paris overtook Rome as a cultural hub in 1789. Schich has argued that the benefit of such a perspective is that 'historians tend to focus in highly specialised areas … but our data allow them to see unexpected correlations between obscure events never considered historically important, and shifts in migration' (Abbot, 2014).

[7] Schich and colleagues address potential biases in their data in the supplementary material of their publication, including 'biographical, temporal, and spatial coverage; curated versus crowd-sourced data; increasing numbers of individuals who are still alive; place aggregation; location name changes and spelling variants; and effects of data set language'.

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Such a framework is generalisable, offering different approaches for large-scale questions. If we conceptualise the individual people within this aggregate model of migration as carriers, we can imagine how researchers could use the framework Schich proffered to chart how patterns of travel and migration disseminate particular materials, technologies, practices, or concepts. The framework also shares important parallels with the work of the GLEAM project at Northeastern University, which has developed an epidemic forecaster that predicts how given diseases might spread by combining data on population, transport networks, the characteristics of disease, and possible responses (like travel restrictions and vaccination efforts). Using this information, the forecaster generates a simulation of disease propagation across the globe. The importance of this project lies in its ability to make informed suggestions about when and where to disseminate vaccines to minimise the impact of potentially devastating epidemics. During the COVID-19 pandemic in 2020 GLEAM provided early insights into the most effective mitigation strategies to slow the global spread of the disease.

The predictive power of network-based models is a new horizon for cultural studies. In general we tend to study artefacts or practices already in existence. Of course we may soon find ways and uses for making predictions about the cultural arena of tomorrow. But prediction is equally important for thinking about existing culture where our data are scarce or missing, reframed as interpolation or computationally assisted historiographic guesswork. An example of this is Shawn Graham and Scott Weingart's research on the problem of 'equifinality' faced on archaeological sites: when a network is uncovered but the final shape of that network could have been reached by many potential means (Graham & Weingart, 2015). In the example studied, the nodes in the network were a series of bricks bearing the stamps of certain producers (carrying information concerning the estate on which the brick was made, the brick maker, the landowner, and the year) at separate sites, which pointed towards a trade network. Using agent-based modelling as a kind of counterfactual laboratory, the authors predicted the various possible processes that might have formed the networks apparent in the archaeological evidence. By comparing the networks that would have been predicted by Peter Fibiger Bang's model of the

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Roman economy, which he calls the 'imperial Bazaar', against those observed in the archaeological materials, Graham and Weingart showed how scholars can employ counterfactual simulations to probe the limits of historical theories. In this case, the methods of prediction are not used to learn something new, but as a formalised thought experiment to better understand the implications of a theorised historical causal explanation. If the simulated results do not match the historical evidence, it is an indication that one's historical explanation needs further articulation.

The foregoing examples, despite the often complex nature or size of the data under consideration, might nevertheless be considered simple in that they constitute unipartite (or unimodal) networks: that is, networks with nodes of just one type. As we have argued elsewhere, however, the kind of networks humanists are interested in are often more complicated: 'We don't just have people connected to people or websites to websites, we've got people connected to institutions to authored works to ideas to whatever else, and we want to know how they all fit together' (Weingart, 2013). This is where we can employ what are known as multipartite (or multimodal) networks.

The flavour network is a bipartite example. Sebastian Ahnert's research with collaborators at the Barabási Lab was inspired by the food-pairing hypothesis proffered by the chef Heston Blumenthal and the food scientist François Benzi: that ingredients will taste good together in a dish if they share chemical flavour compounds. Until their 2011 article, this hypothesis relied on anecdotal rather than large-scale quantitative evidence (Ahn et al., 2011). Thanks to the efforts of food scientists around the world, we now have extensive information available about the many chemical compounds responsible for giving different foods their distinctive smells and tastes. The two node types generated from that data for the bipartite network are the ingredients (such as garlic, chocolate, coffee, basil) and the individual chemical compounds. In order to show the relationships between a particular set of nodes, bipartite networks are usually transformed by a 'one-mode projection'. This means that the resulting network contains nodes of only one of either of the two sets. In this case, a one-mode projection of the ingredients was generated and edges were created between any foods that shared one or more compounds; the edges between those ingredients were

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weighted by the number of compounds they shared. The resulting network was then used to test the food-pairing hypothesis by cross-referring it with a data set of 56,498 recipes. This analysis revealed that Western cuisines show a tendency to use ingredient pairs that share many flavour compounds, supporting Blumenthal and Benzi's hypothesis, but East Asian cuisines tend to avoid compound-sharing ingredients. This work shows how systems and hypotheses that do not seem at first sight to be about networks can be approached productively through that framework.

Where might this kind of approach be useful in the humanities? It is particularly pertinent where one has sources or data containing two different types of entities, and especially when one is a subset of (or somehow contained within) the other. For example, if we think of miscellanies – a collection of literary compositions or pieces by several authors assembled in a volume or book – we could create a bipartite network linking texts (node type one) with the miscellanies that contain them (node type two). So, in our toy network diagram (Figure 9), if text A appears in miscellanies X, Y, and Z, and text B appears in miscellanies Y and Z, then the bipartite network has connections A-X, A-Y, A-Z, B-Y, and B-Z. By then undertaking a one-mode projection of texts connected by miscellanies, we can see which texts appear together in such miscellanies (middle), or conversely, by projecting this as a network of miscellanies connected by shared texts, we can see how closely particular miscellanies are related to one another (bottom). Scaling up this idea, we can begin to see how an analysis of all catalogued Renaissance poetry miscellanies might enable us to understand the early modern poetic universe in different ways. It could tell us which poems most commonly appeared together, it might help us see which manuscripts were most similar in their contents, and it may uncover patterns of scribal circulation.

An example of this kind of bipartite thinking can be found in Richard Jean So and Hoyt Long's analysis of the publication networks of early twentieth-century poets. In their analysis, the two node types are poets and journals, and the fact of publication in the same journal is the edge linking one type of node to another. Their aim was the interrogation of the social dimensions of early twentieth-century poetry production in the United States, Japan, and China. 'The little magazines and independent coterie

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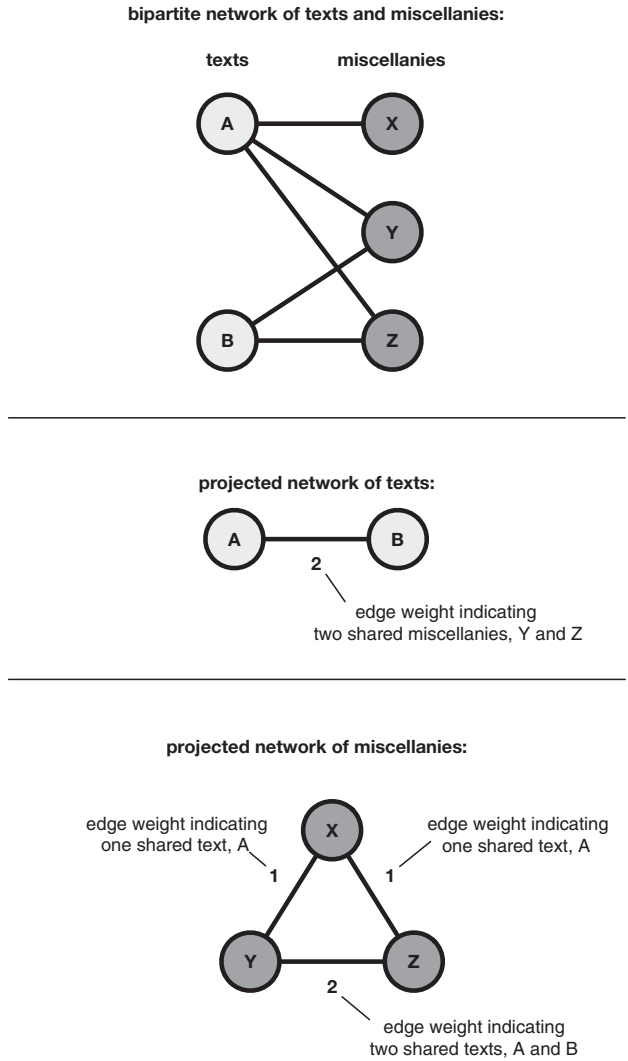


Figure 9 A simple example of a bipartite network made up of nodes and miscellanies (top) projected as a network of texts (middle), and projected as a network of miscellanies (bottom). Diagram by the authors.

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journals of the time were essential mechanisms for organizing creativity, collaboration, and the transnational diffusion of poetic styles' (So & Long, 2013: 157). Their work identifies key brokers in literary circles that include key writers (such as Amy Lowell and Countee Cullen in the network of American poets, 1924–5), and journals (such as *Shishin* in the network of Japanese journals, 1927–9).

For humanities scholars unfamiliar with the network framework, the breadth and inclusivity of the foregoing examples may be surprising. Without the metaphorical and analytic power of networks, it is difficult to understand how the interactions of characters in *Hamlet* are in any way comparable to the dissemination of memes on Facebook, or the chemical compounds shared between food ingredients. What makes them comparable is an intellectual and methodological shift by which we abstract our objects of study into data points that can be entered into a database or spreadsheet. This does not imply some shared property intrinsic to each of the subjects under study, rather it implies the widespread utility of networks as a lens through which to view many aspects of our shared world.

The idea of talking about cultural artefacts as data, metadata, or data sets is in itself a potentially controversial move as it seems reductive, ignoring complexity, ambiguity, and qualitative assessment. Moreover, as Johanna Drucker has argued, the etymology of the word 'data' is itself problematic in the context of the arts and humanities: it comes from the Latin *datum*, which means 'that which is given'. The word 'data' therefore carries with it the meaning of an observer-independent fact which cannot be challenged in itself. Drucker prefers instead to think in terms of 'capta': 'that which has been captured or gathered', a coinage designed to draw attention to the fact that the very act of capturing data is oriented by certain goals, done with specific instruments, and driven by a specific attention to a small part of what could have been captured given different goals and instruments (Drucker, 2011). In other words, it requires decisions and interpretations that introduce subjectivity into the process. Throwing out the word 'data' completely, however, seems unwise given its ubiquity and acceptance across disciplines, especially if one of our aims is to create discourses and practices that break down traditional disciplinary divides. It is perhaps best, therefore, to make sure that scholars from the arts and humanities are not

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held captive to assumptions about what data is, and rather play an active role in applying a self-reflexive and theoretical lens to the act of data creation and capture and its subsequent processing, not just in the humanities and arts, but across the disciplines. Dan Rosenberg's work on the history of data zeros in on the distinctions between data, fact, and evidence. He writes, 'facts are ontological, evidence is epistemological, data is rhetorical,' which means that 'the existence of a datum has been independent of any consideration of corresponding ontological truth' (Rosenberg, 2013: 18). Rosenberg's formulation helps us to see that data are what we determine them to be, but we do need to take responsibility to define them clearly.

Two key issues for network data are those of absence and 'bias' (broadly defined). The concern we hear repeated most often by our colleagues in the arts and humanities is that the incompleteness or complexity of their source material make it unsuitable for network analysis. The concern arises, in part, from a misconception about what 'scientific' data looks like. It is important to remember, however, that almost any network data set, even in the sciences, is incomplete in some sense. In whatever way we define the connection between entities and the entities themselves, we are limited by the scope of available data and by the assumptions made when defining the nature of the connections. While this of course will affect the results of any quantitative analysis, this incompleteness is less of a problem than it may seem at first, for three main reasons.

The first reason is that, particularly in the context of the humanities, the bias in the source data is often itself of interest to the scholar. When we look at cultural phenomena and artefacts, we look at them with particular well-theorised frames. However, for much data we do not have a well-thought-out, critical apparatus to describe the 'how', 'who', and 'what' of the data, so it is not clear whether we are analysing a particular phenomenon, some artefact of the collection and accumulation process, or our own world views. For example, in the case of archives of historic documents, network analysis can help the researcher to better understand the history of that archive by drawing attention to deliberate inclusions or omissions of material, perhaps motivated by a political or religious collection agenda, the confiscation of documents, or even the deliberate destruction or censorship of materials.

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Such events in the history of an archive are detectable precisely because of the effects they have of distorting the results of the quantitative analysis. We might be able to extrapolate similar kinds of lessons for survival or loss of archaeological evidence. And more generally, it could help in the development of a consistent scholarly apparatus for understanding the peculiarities of digital archives and databases. Via source criticism, we still only know in the broadest strokes the implications of their formation and shape for the research we undertake. Articulating humanities network data in these terms, as a perspective rather than a 'bias' – which implies some underlying ground truth – will be a grand challenge for the next generation of researchers in this area.

The second reason is that, even in an incomplete network, the relative importance of nodes and edges according to a given network measurement can still yield meaningful results. Every network data set analysed in network science, be it a mobile phone network, a network between genes, or an online social network, is incomplete or limited by selection. But the limited scope in which the data is collected does not prohibit us from examining the relative importance of nodes or edges in the context of that scope, as long as we are aware of the factors that may shape our results.

The third and most important reason why the incompleteness of networks does not pose an insurmountable problem is that the results of the quantitative analysis do not serve as a final result, but as a starting point for further, more detailed inquiry in the vein of traditional scholarship in the arts and humanities. In Chapter 6, we provide a guide through what this might mean practically for the research process.

The message that networks do not need to be complete should not be used to discount the problem of inconsistent sampling or poor data gathering. But network analysis can be part of the answer if it is employed as a tool for characterising the contours of a given archive or corpus, thereby helping to show how the data with which we work so often reinscribes structural bias. This raises an important issue when understanding and constructing cultural objects as data. The big-data approach to culture offers many opportunities, as discussed earlier, but with it comes a duty of care that is at once technical and ethical (Weingart, 2014a). These two aspects are intimately intertwined.

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The technical duty of care is often called 'data cleaning', a term used to describe the process of detecting and correcting (or removing) corrupt or inaccurate elements in a given data set. Issues within a single data set or database might be created by misspellings during data entry, missing information, or other invalid data. The term, however, is not without problems, as has been discussed in the provocatively titled piece 'Against Cleaning' (Rawson & Muñoz, 2019). Rawson and Muñoz's point is that the metaphor of cleaning is problematic because it elides the burden of labour and intellectual contribution that it actually represents. They contend that: 'Data cleaning is a consequential step in the research process that we often make opaque by the way we talk about it. The phrase "data cleaning" is a stand-in for longer and more precise descriptions of what people are doing in the initial phases of data-intensive research.' The lack of discussion around such practices, they argue, has increased mistrust of quantitative approaches in the arts and humanities. The only way to deal with this is to begin talking about the labour of cleaning and to communicate its significance as an intellectual contribution.

One example is the preparation Ruth and Sebastian Ahnert undertook for the Tudor Networks of Power project. Their primary data set was a digitised archive, State Papers Online (SPO), which combines scans of the British State Papers archive with the Calendar descriptions of their contents. It provides metadata for each document that, in the case of the letters on which they were working (numbering circa 132,000 items for the Tudor period), contains valuable relational information including: name of sender, name of recipient, date of composition, place of writing, unique document identifiers, and a content description. The author, recipient, and place fields in particular needed cleaning. In the case of the author and recipient fields, the names needed disambiguation and de-duplication, for a number of reasons: variant spellings of early modern names, letters addressed to a titular office rather than a named individual (e.g. the Archbishop of Canterbury), changing office holders, changing titles over a person's lifetime as they accrued honours and offices, and women's names changing due to marriage. The complexity of the sender and recipient metadata fields meant that although 37,101 unique name entities were initially extracted, there were in fact only 20,656 unique correspondents. Place names had to be dealt with in a similar

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way and geo-coordinates added. This effort led to the development of two custom data-cleaning tools. In addition, the curation of those data fields involved eighteen months of work (Hyvönen et al., 2019; Ahnert & Ahnert, forthcoming: chapter 1). This is a considerable workload that needed to be undertaken before network analysis could even be contemplated and the results of quantitative analysis could be trusted. This work should not just be considered a prerequisite for network analysis or other quantitative approaches, however, but a scholarly object in its own right: a data set other researchers can put to many uses. A cultural shift in the humanities is still required before such work is given the scholarly credit it deserves.

A vital by-product of this labour is that during the process researchers become intimately acquainted with the shape of their data, its strengths, weaknesses, and biases. This is where the ethical duty of care comes in. Lauren F. Klein's MLA paper on 'Distant Reading after Moretti' argued that we need to look at our data to understand why 'distant reading does not deal well with gender, or with sexuality, or with race.' She suggests that we need data sets that:

> perform the work of recovery or resistance. An example: the corpus created by the Colored Conventions Project, which seeks to recover and aggregate evidence that documents the Colored Conventions of the nineteenth-century United States; these were organizing meetings in which Black Americans, both fugitive and free, came together to strategize about how to achieve social and legal justice. By making this corpus available for others to download, the CCP opens up the project of distant reading to texts beyond quote 'representative' samples, which tend to reproduce the same inequities of representation that affect our cultural record as a whole. (Klein, 2018)

Attending to the biases created in the creation of the archives with which we work allows us to address them. Networks can be part of that process, helping us to describe the data we have, and therefore allowing us to see where gaps and skews are present and to think critically about the ways in which they may be addressed.

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By thinking about the culture around us as data we therefore not only open up opportunities for new analytical processes, we also identify an important way in which arts and humanities scholars can contribute critically to the ways we theorise the construction of data sets. The ways that this community is accustomed to thinking about these objects of study offers a different perspective on the status of data, its capture, preparation, shape, strengths and weakness, as well as the duty of care required to ensure it is fit for purpose. As the next chapter argues, one of the ways we can interrogate our data is through the lens of visualisation.

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4 Visual Networks

Visual networks are compelling, but are they effective? The conventional network graph of node and edge (points connected by lines) makes it possible to convey a tremendous amount of information all at once, in one view. Networks express an internal logic of relationships between entities that is inherently intuitive. But they also lack an explicit external spatial referent, whether the latitude and longitude of cartography, the scale and sequence of a timeline, or the categories and measures that mark the x-y axis of a statistical graph. They can represent relationships at human visible scale, or scale to dizzying, visually indecipherable complexity. The unbounded rhizomatic structure of the network has a malleable, infinitely re-orderable form that communicates in a way that is distinct from almost all other diagrams of data. This chapter addresses how that malleability can be harnessed for research and communication.

The sketch presented in Figure 10 is one of many drafts leading up to Mark Lombardi's work, 'BCCI-ICIC & FAB, 1971–91 (4th Version)', with which we opened this book. It is a rough, incomplete, marked-up draft with nodes outlined in blue, connections crossed out, and empty spaces awaiting more information. A comparison between this draft and the final version gives us some insight into the evolution of Lombardi's thinking. Externalising information and organising ideas spatially is not just a mode of presentation, it is an integral part of Lombardi's research into these complex financial systems. The story he finally shares has been refined through the process of sketching, editing, and redrawing. But it is not merely a cleaned-up version of the draft. It demonstrates Lombardi's command of visual rhetoric to lay out his argument for public consumption; to convince and persuade with his choice of colour, the quality of the line, the careful annotation, the apparent precision, the internal structure, and the overall shape. The fragments of evidence Lombardi originally collected on three-by-five notecards and arranged in notes and sketches do not speak for themselves; they are selected, edited, and enriched, using visualisation at stages throughout the process as a cognitive aide, a modelling activity, an argument, and a work of art in turn (see Tversky & Suwa, 2009 and Tversky, 2014). It is an iterative act of knowledge production that if executed in prose rather than in drawing would

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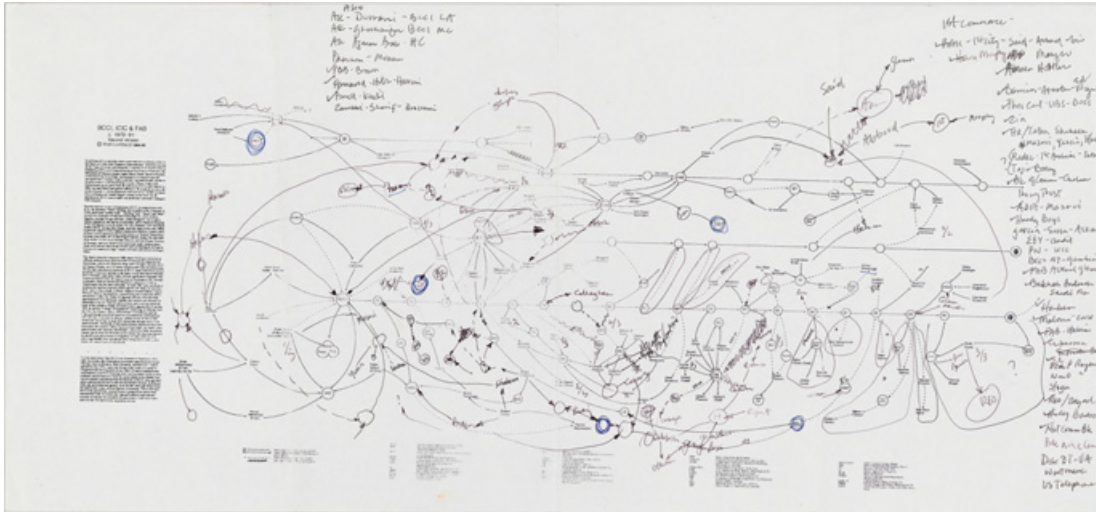


Figure 10 Lombardi, Mark (1951–2000): Untitled from the series BCCI, ICIC & FAB, 1996. Pen and ink and electrophotographic print on paper. New York, Whitney Museum of American Art. © 2019. Digital image. Whitney Museum of American Art / Licensed by Scala. figure 1

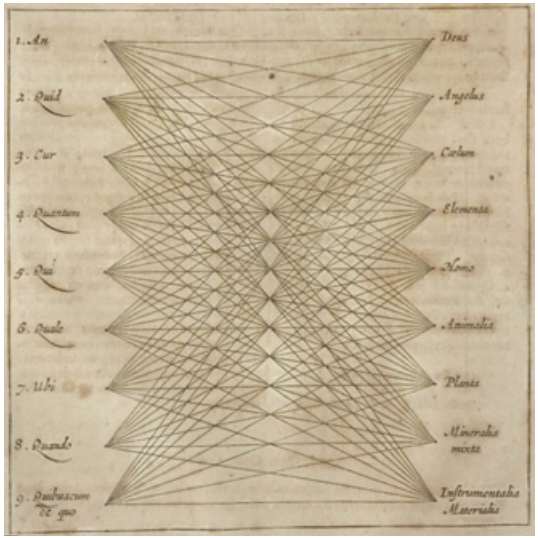
be very familiar to humanities scholars. This familiarity is an important point of reference when comparing hand-drawn visual networks like Lombardi's to those drawn with computer graphics software. Ben Fry, for example, points to Lombardi's work for lessons in a humanist view of data that uses visualisation as a story-telling medium requiring significant editing (Fry, 2016). Our contention is that handled in the right way, network visualisation can be both an important means of knowledge production and a powerful rhetorical tool.

The role of the visual in the production of knowledge is the thesis of Johanna Drucker's *Graphesis*. She opens the book with a comparison of two network visualisations: Athanasius Kircher's *Ars magna sciendi* (1669) and Barrett Lyon's 'Web Traffic Visualisation'. Both are networks depicted as nodes and edges (see Figure 11). Kircher returns us to the diagrammatic approaches to knowledge discussed in Chapter 2. It is a development of Ramon Llull's system published in his *Ars generalis ultima* or *Ars magna*: a method of combining religious and philosophical attributes selected from a number of lists, designed to engage Muslims in debate and win them to the Christian faith. Kircher's diagram forms a complete bipartite graph, where every node of the first set is connected to every node of the second. The Lyon visualisation, part of his Opte Project, is a computer-generated network graph intended to map the Internet. Nodes represent individual servers and hosts, and edges represent the fibre, copper, or other connections between them. Drucker's purpose in juxtaposing these two graphs is to draw our attention to their relationship to information. Kircher's work, she argues, '*produces the knowledge it draws*': the effect of combination is to generate insights. By comparison, Lyon's network visualisation of web traffic 'only displays information'. This distinction is an important one when thinking about how we employ visualisation in our work. Off-the-shelf graphics packages are employed in research as a means both of generating insights and displaying observed phenomena. However, the word 'display' is a potentially misleading term. As Drucker puts it, certain visualisations 'act as if they are just showing us *what is*, but in actuality, they are *arguments made in graphical form*' (Drucker, 2014). We need to recognise the rhetorical power of visualisation if we are to remain alert to the hidden agendas of the graphical forms we encounter.

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(a)



(b)

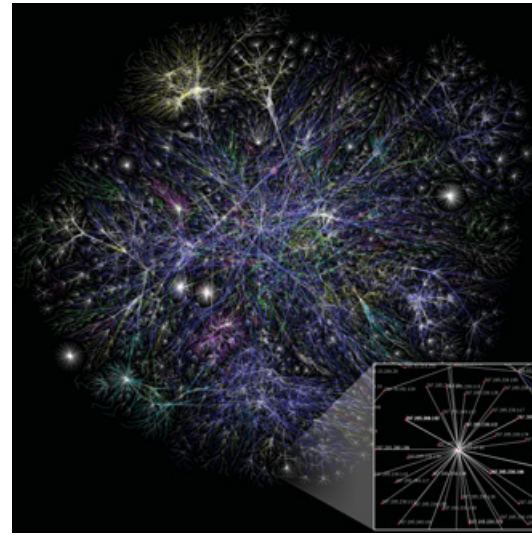


Figure 11 On the left, a combinatorial diagram entitled *Typus universalis, omnibus de quacunque re proposita questionibus formandis, aptus*, in Athanasius Kircher, *Ars magna sciendi* (Janssonius a Waesberg, 1669). Digital image. Max Planck Institute for the History of Science Library. CC BY SA 3.0. On the right, a visualisation from the Opte Project of the various routes through a portion of the Internet in 2005. Digital image. Wikimedia Commons. CC BY 2.5.

We can begin to understand how arguments are made in graphical form by looking at a set of diagrams based on the networks chapter of Jacques Bertin's *The Semiology of Graphics* (Figure 12). Bertin defines a general theory of graphics as separate from both figurative representation and mathematics. While Bertin employs some problematic assumptions about both the certainty of data and the 'monosemic' system of graphical rhetoric, his work is nevertheless a crucial study for our thinking about different insights, and consequently the arguments that can be made through different diagrammatic layouts of network data.

The network, according to Bertin, is represented by three components: the line, the point, and the area. He specifies a number of possible layouts making use of those components including rectilinear, circular, irregular arrangement, and regular arrangement (see Figure 12). Each combination of components and arrangements highlights different aspects of the data and contributes to a different argument about the underlying data. The tabular view provides the basic source–target pairs. The 'irregular arrangement' is the familiar, decentred layout most often associated with visual networks. It does not spatially prioritise any node and yet, with a network this small, we can quickly assess that A, B, C, and E are more connected than the other nodes. The 'regular arrangement' explicitly sets A as the root node in a tree-like hierarchy. The top-to-bottom order of the horizontal planes of alignment is clearly intentional and insists on a particular reading of the graph, even if the significance is not made explicit. And whereas the 'regular arrangement' gives prominence to A, the circular layout emphasises the node with the most connections. The lines radiating out from C give it a visually significant role in the network even though all the nodes are aligned on a periphery. The rectilinear layout suggests a sequence to the connections and makes clear, with this particular data set, that E connects more nodes along the chain than any other node (which, as we see in Chapter 5, would mean it has the highest betweenness centrality). The parallel alignment diagram and the matrix reveal directionality and absence more explicitly than the network graphs because they both draw corresponding connections between two fixed planes. Since these two layouts accommodate all possible connections, they draw our attention to where there are

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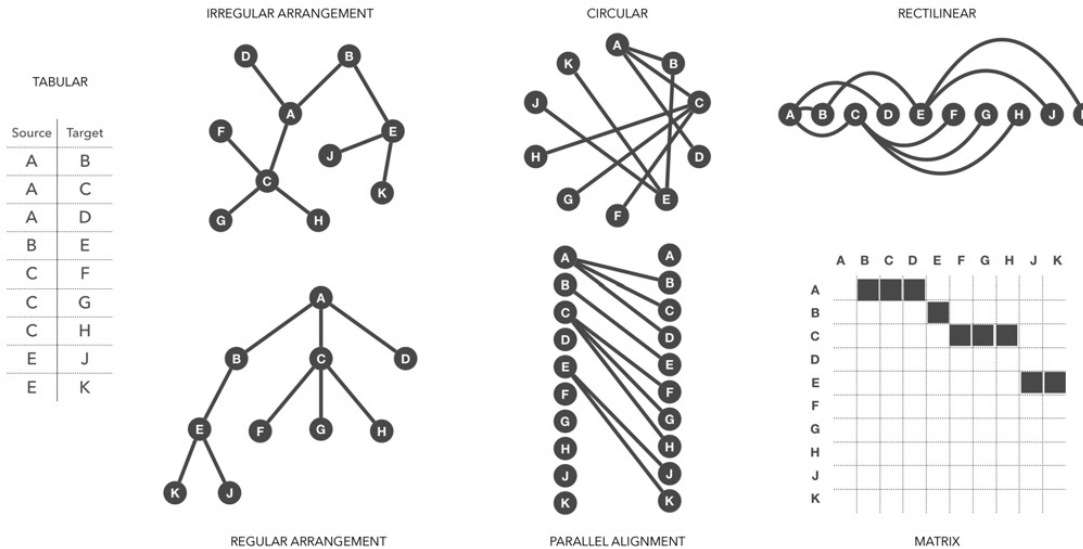


Figure 12 This figure shows a number of different ways to present the same network data. On the left is source–target pairs in a two-column table. The regular and irregular arrangement, circular, and rectilinear are all network layouts in Bertin's system. He distinguishes those node-link network graphs from the diagrams: parallel alignment and matrix. Diagram by the authors.

none. Even with this simple overview, it quickly becomes apparent how much the arrangement of nodes influences our reading of a network.

What is useful about Bertin's work is that it gives us a point of reference for reading the rhetoric encoded in design decisions. For example, if we return to Lombardi, we can see Lombardi's drawings align with Bertin's rules and extend them in unique ways. Lombardi's networks are hand drawn with a fine black line on white paper, sometimes accented with red, with descriptive labels and notes. He distinguishes six different types of connection between nodes: 1) influence or control in one direction, 2) mutual connection, 3) financial connection, 4) sale or transfer of assets, 5) a blocked or incomplete transfer, and 6) the sale or spin-off of a property. Moreover, Lombardi uses the rectilinear arrangement of nodes to add a temporal dimension to the visual network. In some cases, instead of a one-to-one relationship between node and entity, an organisation (usually a financial institution) is represented as a horizontal line with a beginning and an end. These are the primary entities, depicted as chronologies rather than singular nodes. The differences in the quality of the lines (solid, dashed, broken, twisted) are difficult to see at a distance, making the scale of his work an important part of the experience of it (the final work is 129.9 cm x 349.9 cm). Stepping back from the work, it is possible to take it all in as a complete system. The labels and notes are not legible at that distance, though the lines and vertices are clear. Even if the temporality is not explicitly visible, the horizontal lines add another layered dimension to the connectedness. A distant reading complements the close reading. Standing next to the work, the different types of connection and the handwritten labels and notes are easy to see. It encourages the reader to follow paths through the network, shifting their own position in relation to the artwork.

Researchers have made attempts to automate Lombardi's layout, to capture it as a reproducible algorithmic process. Indeed, there is a significant body of work in graph drawing studies called 'Lombardi drawings' (e.g. see Duncan et al., 2010). Such attempts, however, ignore the real value of his drawings and what is at stake in their creation. Lombardi was interested in evidence-based *visual* argument (the 'narrative structure') about specific individuals and organisations engaged in particular types of relationship

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over time. And his sketching process, leading up to the final presented work, captures the process of formulating that argument. Conversely, software programs for visualisation use predefined graphics libraries where much of the decision-making is pre-baked into the tool. They therefore take out of the user's hands the experience of the roughly drawn sketch as part of a generative and exploratory thought process. Instead, it jumps right into the visually precise presentation that we associate with the argument. It is not only that computer graphics look precise, computer graphics software is encoded with assumptions about graphical meaning rooted in Bertin's theory of graphics as monosemic and unambiguous.

Just because visualisation tools present results to us in ways that can be understood as deterministic does not mean we need to use them that way. But we do need to be aware of the assumptions encoded in the tools we use so that we can bend them to our own needs. What follows is a comparison of two network visualisation packages often used in the humanities: Palladio and Gephi. Palladio uses interaction, filtering, and complementary visualisation modes to facilitate the exploration of multidimensional data sets. Gephi uses dynamic simulation to apply layout algorithms to generate graph network visualisations. The comparison is guided by what we expect to gain from the visualisation of data. Isabel Meirelles uses the words 'descriptive' and 'argumentative' to distinguish between visualisation for the exploration of data and visualisation for the communication of findings. She is careful to note that these are not either/or categories, but two poles of a continuum. Descriptive visualisation, Meirelles writes, is a reflection of direct mappings of the data – the who, what, where, and when – whereas argumentative visualisation tends to the more speculative (Meirelles, 2018). This approach to visualisation foregrounds our relationship to the data to be visualised. Lombardi's BCCI project and some of his collected data are the common point of reference in our comparison to help reveal the advantages and potential pitfalls of using software-generated visual networks in the research process.

Palladio is a toolset rooted in an historical-cultural orientation to network analysis that is more art than science. It is the result of experimentation with custom-made visualisations designed to answer specific historical questions

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about the shape of political, social, and intellectual communities in the early modern period (Edelstein et al., 2017; Coleman, 2020). The techniques deemed successful in those experiments were distilled and combined into one suite of tools. The result is a basic force-directed bipartite network graph embedded within an interactive and interchangeable set of data views that includes a geospatial view, a tabular view, and filtering on temporal and categorical dimensions. The interactive tools are designed to work together to help reveal many types of connectedness by facilitating multiple different points of view combined with faceting and filtering on different dimensions of the data. It is assumed that no one view is complete, but that a more complete picture is understood as much by seeing what is missing as by seeing the extant data (see Rendgen & Weidemann, 2012: 155; Uboldi et al., 2013).

An exploration of Lombardi's data set with Palladio, for example, might begin with a map of the known locations. The text printed on the side of Lombardi's *Untitled* sketch (Figure 10) indicates that BCCI was the brainchild of a Pakistani banker based in Karachi. The bank was controlled from Abu Dhabi, was incorporated in Luxembourg, was operated from London, Geneva, and the Cayman Islands, and was involved in joint ventures with local banks in Iran, Oman, France, and Switzerland. Since BCCI was, according to Lombardi, the legitimate front for the shadowy ICIC, examining the locations of individuals he believed were involved could reveal potential connections that might spur further research. Adding the time span filter in Palladio would make the data Lombardi has about when the entities formed and dissolved, cross-referenced with the timeline of specific dates when particular individuals met or financial transactions were recorded, available to filter, making it possible to quickly and easily zero in on which activities are taking place where and at what time. And since Lombardi seems to have more information about the relationships between people and organisations than about locations, he can look at the same filtered subset of data in a graph view to reveal connections between individuals and organisations at those moments in time.

This example demonstrates how Palladio can be used for sketching, in the sense of exploring, or thinking through data as a cognitive aid. It aids cognition by combining visualisations with the counting of values and categories that can easily be grouped and regrouped, arranged and

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rearranged. The faceting and filtering combines with the visual description to enable a subtle interrogation of heterogeneous, multidimensional data sets by concept and duration rather than purely numeric values. It was designed to be used with historical and archival data, with the assumption of incompleteness and uncertainty in the sources. Successful use of Palladio therefore requires the researcher to make inferences based on the visualised data, but it leaves space for the expert to fill in the gaps in the many cases where their knowledge of the subject extends beyond the data collected. It minimises automatic animation in favour of explicitly selected actions, to encourage reflection as views are reoriented and data is manipulated. The application is there not to run an analytical model but to facilitate modelling, to create space for assessment and imagination.

Gephi, by comparison, was developed in the context of the interdisciplinary *e-Diasporas* project to analyse online migrant communities (Diminescu, 2012). The underlying data, referred to as an *e-diaspora corpus*, is a list of diasporic websites selected by the project authors. The tool reflects the project's interest in the power of network science and the desire to visually expose network topology. Its creators describe it as the Photoshop of network graphs and it uses computer simulations to show layout algorithms acting on network data (Bastian, Heymann, & Jacomy, 2009). The force-directed layouts are a class of algorithms used frequently in Gephi and the visual form with which readers will be most familiar because of their increasing ubiquity through high-volume use of Gephi and similar tools. In a force-directed layout, each element of the network is modelled as though guided by unseen forces, much as in the physical world: gravity, electromagnetic repulsion, material elasticity. Nodes may be thought of as electrons on a two-dimensional plane, forcing each other apart as they approach one another, and the edges may be conceived of as springs physically anchored between two nodes, drawing them together even as they repulsively push away. The plane has no meaning; there are no axes defining categories against which to measure node position. But as we are already aware, we do come to these resulting images with a set of assumptions about meaning. Gephi also allows users to apply a whole range of quantitative measures (discussed further in Chapter 5) and then colour or resize visual elements according to those attributes.

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An analysis of Lombardi's data set with Gephi would make it possible to automatically generate a calculated layout based on the connectedness of nodes in his network. Layout models embedded in a software application are extremely powerful. It makes it easy to quickly apply a given model to lots and lots of data. However, by making it so easy to apply multiple different layouts to the same data set, small data problems can be overlooked and amplified. In the scientific context, where network analysis is based on equations and resulting measures, computer-generated graphics like those produced with Gephi have come under scrutiny for unnecessarily introducing ambiguity into a reading of network topology. The argument is that spatial arrangements, colours, shapes, and labels complicate measurements that are more clearly presented numerically. But the source of confusion in network graphics resulting from powerful visualisation tools like Gephi may have more to do with the fact that visualisation is too often learned rather than taught. Too often users allow the software to determine the layout or become enamoured of the visualisation and animation, losing sight of the intent to accomplish something on the spectrum of description to argument.

Tommaso Venturini, Mathieu Jacomy, and Pablo Jensen have begun to address this challenge by explicitly defining the strengths and weaknesses of what they have called visual network analysis (VNA) and providing guidelines for how to make effective use of algorithmic layouts to expose the topological structure of large networks (Venturini, Jacomy, & Jensen, 2019). The diagrammatic network layouts Bertin offered – which provide visual clarity through either symmetry or minimising overlapping connections – break down when networks reach the size of several hundred or thousands of nodes. The argument underlying VNA is that understanding the conceptual and mathematical underpinnings of the layout algorithms and choosing appropriate settings can result in effective, legible graph spatialisation. Even though the specific position of each node on the plane is not relevant, the density (or lack of density) of nodes is. And though the structure of the visualised graph does not strictly correlate with the underlying mathematical calculations of the network, the structure that emerges through polarisation of nodes, even in a very large network, functions effectively as exploratory data analysis (Tukey, 1977). Selectively highlighting variables in the data reveal patterns. Nodes can be sized according to degree measures (see Chapter 5);

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brightness can be applied to continuous numerical variables and hue for categorical variables. And just as with the experience of Lombardi's drawing, here closeness and distance are both revealing: zooming in and out in a Gephi visualisation makes global and local patterns visible. Even if the end goal is not a statistical reading of the network, the statistical readings available through Gephi provide intellectual prompts to examine your data in new ways.

Lombardi's work, Palladio, and Gephi represent three different kinds of visual engagement with data that require and deliver different levels of artistry, control, and quantitative engagement, but all of them encourage an important process of exploration. Network visualisations are often critiqued as either overly reductive (in terms of content and context) or absurdly illegible (in terms of visual complexity), yet they bring an immediacy to our perception of information. Graphics as a system for the creation and presentation of knowledge requires our attention. Graphics can, and often are, used throughout a research process from sketch drawings that help us build intuitions about connectivity, to descriptive data visualisation, to visual arguments in a final publication. Though data analysis is widely considered synonymous with particular forms of visualisation these days, charts and graphs were not widely accepted until they were formalised in the mid-twentieth century when computing and, specifically, computational information processing was on the rise. Statisticians Jacques Bertin and John Tukey, whose careers spanned the latter half of the twentieth century, were both pioneering figures who solidified the role of visualisation in providing insights that would otherwise be missed in mathematical data analysis. Bertin anticipated that the automation computers provide in calculation, combined with the ability to transform and reorder via computer display, would enable visualisation of much larger data sets. He seemed confident that computer graphics would be the apotheosis of his goal to produce 'the simplest and thus most communicable image'. However, one could argue that the ideal Bertin imagined has gotten away from us.

Cultural phenomena that Bertin could not have anticipated have radically changed our relationship to data graphics. Our day-to-day familiarity with mapping services is a case in point. We not only accept but rely upon their cartographic conventions to reduce the complexity of the landscape in order to serve a particular purpose: wayfinding. We navigate from point to

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point, implicitly accepting the abstraction of the map because we have our embodied experience of walking, riding, and driving streets to fill in the gaps. We know that travel between two nodes is rich and layered; it is never a straight line. We have learned to reframe space in scale from a global view to the city streets with longitude and latitude and familiar geography as a guide. We trust these constantly shifting maps perhaps because we are so aware of their limitations or because we rely on their convenience. As a result, we allow ourselves to be led by them, unaware of or tacitly accepting their encoded intention. The intervention of software as an unseen hand in the construction of visualisations presents new challenges to our reading of them. When the direct connection between the author and the output is not evident in the form, we can be seduced into experiencing visualisation as objective. This is even more likely when the map layer is removed and what is left is a visualised network graph without a physical referent.

Spatialisation of nodes and links, vectors and vertices, without reference to scale or familiar geospatial markers is unsettling and confrontational. And yet, eschewing the Cartesian hand-holds of the map reveals patterns and movement that would otherwise be obscured or overshadowed by the expectations we bring to the geospatial framing of data. It focuses our attention differently – from topography to the line and its inherent relationships. The points, lines, and areas by themselves can be disorienting and challenge our assumptions about what is significant in the data. At the same time, our relationship to data has changed radically in this century. Not only is there much more of it, it is much more personal. As a result, we are better able to see the human-scale biases and assumptions behind the abstract and inherently fragmented visualisation of networks. Just as networks are always metaphorical, visual networks are as much about schema as they are a set of relationships and measurements. These opportunities and challenges of the network have been seized upon by artists, scholars, and, more recently, the developers of visualisation tools and libraries. The need to concretise the abstract has given way to the desire to communicate expressively now that we have extremely powerful tools with which to fulfil that desire. The challenge is to redefine the role of graphics in network analysis for the future of network science, design, and humanistic inquiry. Dan Rosenberg, for example, posits a different analytic goal of

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visualisation: 'to complicate rather than simplify, to open multiple avenues of inquiry, and, most importantly, to challenge the stability of underlying data, in fact or in principle' (Rosenberg, 2016).

The fact that dense and complex networks resemble neural connections in the brain, rhizomatic weed roots pulled from the ground, twenty-four hours of airline flight, or the exchange of information over the Internet, communicates something about the very real complexity we experience in our everyday lives at many different scales, which we need to interrogate. We propose that visualisation is not better than narrative argument or mathematical equations for communicating ideas, but that it provides an additional means of producing, exploring, and analysing information that has proven value in both the liberal arts and the sciences. When working with data, visualisation can help reveal gaps that narrative-based research practices do not expose, or uncover qualities in data sets that can easily be overlooked when we focus only on numbers and calculations. Moreover, it is rhetorically powerful. Visualisations of networks are unsettling because the graphical language used to produce them is not, in fact, precise. For these reasons, we argue that visualisation has a challenging and provocative role to play in the development of network approaches to culture, a role that needs to be fully integrated into an iterative and self-reflexive research process. In Part III, we outline the manoeuvres that make up that process.

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Part III Manoeuvres

Network analysis provides a set of manoeuvres: intellectual manoeuvres that refigure cultural objects in our minds as abstract systems of nodes and edges; mechanical manoeuvres that structure data and that navigate input versus output; and manoeuvres between a landscape of abstraction and research questions that are steeped in contextual information. These manoeuvres dismantle the binaries between distant and close reading, quantitative and qualitative analysis, the nomothetic and idiographic, and, in so doing, create new norms of practice and inquiry.

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